

Practical Homework P4: potter KB [Joelle Dizon]

The Team:

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Step #1: Importing necessary modules.

The block below imports a necessary module and checks if you are in the right directory. If you do not see a list of python files when you run the block, please make sure you have uploaded the zip file and unzipped it.

```
In [1]: import matplotlib.pyplot as plt
```

```
%pwd  
%ls
```

all_facts.txt	nnunifier.py
atomgenerator.py	potter_model.pt
autoencoder.py	potter.txt
basictypes.py	__pycache__ /
chainbased.py	randomKB.txt
'CSE 327 P4.ipynb'	reasoner.py
embedmodel.py	rKB_model.pth
evaluate.py	std-46-110-3-50.csv
gameofthrones.txt	termwalk.py
gen_queries.py	test_queries.txt
guided_loss-unification46-110-3-50.png	training_and_validation_loss.png
helpers /	training_loss-10-100-4-50.png
kbencoder.py	training_loss-45-106-2-50.png
kbparser.py	training_loss-46-110-3-50.png
knowledgebase.py	train_queries.txt
ming-46-110-3-50.csv	triplets.csv
mr_back_reasoner.py	vocab.pkl
mr_train_examples.csv	vocab.py
nnreasoner.py	

Step #2: Prepare the potter KB

Creates the file:

- vocab.pkl - Indexes the full set of predicates, constants and variables used by the knowledge base.

```
In [11]: !python kbencoder.py --kb_path potter.txt --vocab_from_kb --save_vocab
```

Creating vocabulary from last potter.txt knowledge base

Step #3: Train the neural net

Train a neural net to convert first-order logic atoms into meaningful numeric vectors. We call these vectors "embeddings," and the learned model the embedding model. The code creates random atoms and then uses a mechanism known as triplet loss to place unifying atoms closer together.

Creates the files:

- rKB_model.pth - Saves the learned weights for the embedding model
- training_loss-45-106-2-50.png - A plot of the training loss after each epoch. This file's name depends on the vocabulary of your KB: the four numbers are number of predicates, number of constants, max arity of the predicates, and the embedding size used. If you changed the KB, you'll need to look up the name in your drive and update the code block that displays this file.

```
In [4]: !python kbencoder.py --kb_path potter.txt --train_unification_model
```

Using cuda device

Uni model embed size: 50

```
[██████████] 3500/3500 Anchors generated  
[██████████] 70000/70000 Triplets generated  
[██████████] 18435/70000 Triplet encodings
```

IOPub data rate exceeded.

The Jupyter server will temporarily stop sending output to the client in order to avoid crashing it.

To change this limit, set the config variable
`--ServerApp.iopub\_data\_rate\_limit`.

Current values:

ServerApp.iopub_data_rate_limit=1000000.0 (bytes/sec)

ServerApp.rate_limit_window=3.0 (secs)

```
[██████████] 40954/70000 Triplet encodings
```

IOPub data rate exceeded.

The Jupyter server will temporarily stop sending output to the client in order to avoid crashing it.

To change this limit, set the config variable
`--ServerApp.iopub\_data\_rate\_limit`.

Current values:

ServerApp.iopub_data_rate_limit=1000000.0 (bytes/sec)

ServerApp.rate_limit_window=3.0 (secs)

```
[██████████] 50829/70000 Triplet encodings
```

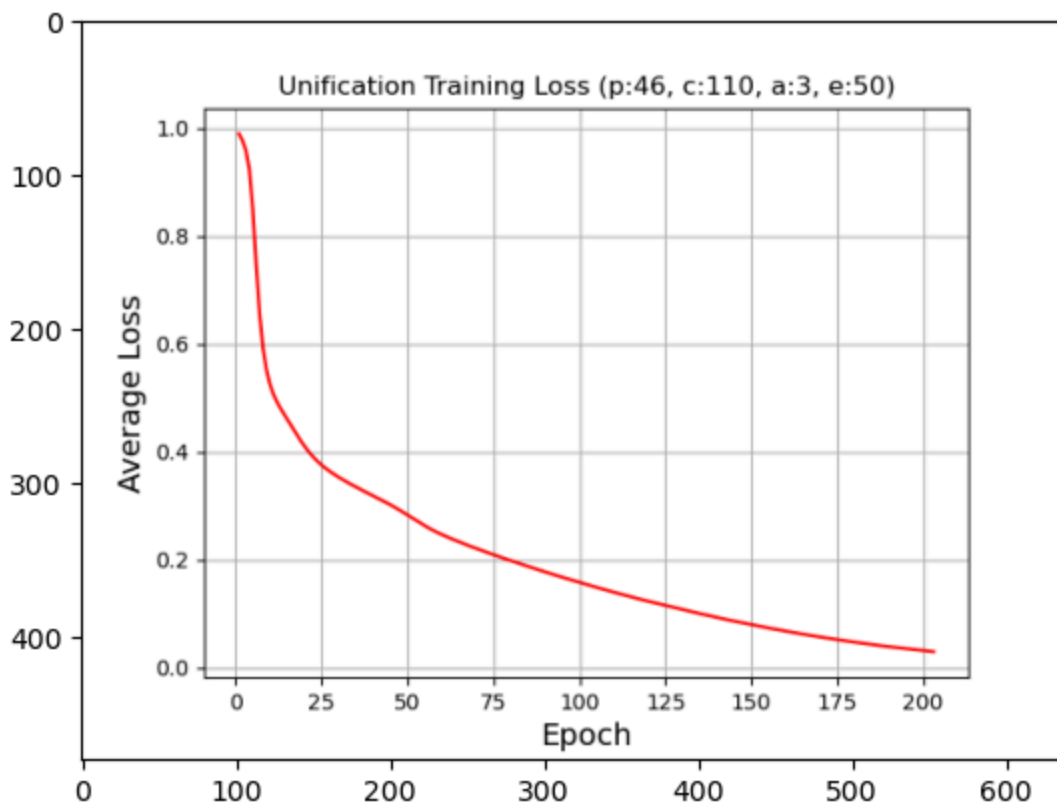
IOPub data rate exceeded.
The Jupyter server will temporarily stop sending output
to the client in order to avoid crashing it.
To change this limit, set the config variable
`--ServerApp.iopub\_data\_rate\_limit`.

Current values:
ServerApp.iopub_data_rate_limit=1000000.0 (bytes/sec)
ServerApp.rate_limit_window=3.0 (secs)

[██████████] 70000/70000 Triplet encodings
Early stopping triggered-----] 40.5% Epoch | [Epoch 203/500, Trainin
g Loss: 0.0292, Validation Loss: 0.3410]
Saved model.
Saved plot.
unification model generated

```
In [6]: img = plt.imread('training_loss-46-110-3-50.png')  
plt.imshow(img)
```

Out[6]: <matplotlib.image.AxesImage at 0x71d6305c0890>



Step #4: Generate set of train & test queries

Generate a set of train and test queries from your knowledge base. This will also generate the deductive closure (all facts that can be inferred) of your KB using forward-chaining.

Produces the files:

- all_facts.txt - A list of all facts that can be inferred from the KB. This is used to generate queries that are guaranteed to have an answer.
- train_queries.txt - A set of training queries where each will have at least one answer.
- test_queries.txt - A set of test queries where each will have at least one answer.

```
In [7]: !python kbencoder.py --kb_path potter.txt -g
```

```
Generating facts list...
Forward chaining depth: 1
New facts: 272
Forward chaining depth: 2
New facts: 137
Forward chaining depth: 3
New facts: 16
Forward chaining depth: 4

Facts list generated
425 total facts. Max depth = 3

Number of queries: 200
100 training queries generated
100 testing queries generated
```

Step #5: Create a training set for the scoring model

Create a training set for the scoring model. The examples will consist of goal, rule, score pairs, where a score of 1 means the rule can solve the goal, and 0 means it cannot. It runs the full set of training queries with a randomized backward-chaining process. This step can sometimes take a long time and is not able to take advantage of a GPU (with the gameofthrones KB it took ~45 minutes).

Creates a file:

- mr_train_examples.csv - Contains a line for each goal/rule/score example.

```
In [8]: !python kbencoder.py --kb_path potter.txt -p
```

running training queries...

Negative facts will be generated

1: [enemy(_EX4, seamus_finnigan)]

min depth: 3

Nodes: 15

Answers: 6

2: [ancestor(P, neville_longbottom), descendant(neville_longbottom, P)]

Restart...

min depth: 2

Nodes: 1640

Answers: 2

3: [magical_pet(P, fluffy), male(P)]

min depth: 5

Nodes: 6

Answers: 1

4: [enemy(P, harry_potter), professor(P), enemy(P, X), enemy(narcissa_malfoy, X)]

[-----] 0.0% to max nodes (296120 nps)Mismatched predicates: professor(P) with member_of(harry_potter, death_eaters)

min depth: 1

Nodes: 28

Answers: 17

5: [parent(X, P), ancestor(arthur_weasley, P), hates(Y, X), enemy(minerva_mcggonagall, Y), parent(X, rose_weasley)]

min depth: 1

Nodes: 34

Answers: 33

6: [hates(draco_malfoy, _EX0)]

min depth: 4

Nodes: 20

Answers: 10

7: [professor(P), male(P)]

min depth: 1

Nodes: 14

Answers: 13

8: [male(P), descendant(P, frank_longbottom)]

min depth: 1

Nodes: 44

Answers: 43

9: [enemy(remus_lupin, _EX1)]

min depth: 3

Nodes: 30

Answers: 9

10: [in_gryffindor(_EX1)]

min depth: 2

Nodes: 20

Answers: 18

11: [enemy(harry_potter, _EX1)]
min depth: 3
Nodes: 39
Answers: 11

12: [male(P)]
min depth: 1
Nodes: 44
Answers: 43

13: [enemy(alastor_moody, severus_snape)]
min depth: 3
Nodes: 5
Answers: 1

14: [male(P), professor(P)]
min depth: 1
Nodes: 44
Answers: 43

15: [descendant(X, P), ancestor(P, X), enemy(draco_malfoy, X)]
Restart...
min depth: 3
Nodes: 4595
Answers: 33

16: [in_hufflepuff(_EX5)]
min depth: 2
Nodes: 6
Answers: 4

17: [enemy(X, P), female(P), female(X), married(Y, P), descendant(hugo_weasley, Y)]
min depth: 1
Nodes: 219
Answers: 199

18: [descendant(_EX5, ginny_weasley)]
Restart...
min depth: 3
Nodes: 16
Answers: 3

19: [enemy(P, peter_pettigrew)]
min depth: 3
Nodes: 14
Answers: 10

20: [in_slytherin(_EX2)]
min depth: 2
Nodes: 13
Answers: 11

21: [subject(P)]
min depth: 1

Nodes: 12
Answers: 11

22: [wizard(arthur_weasley)]
min depth: 3
Nodes: 7
Answers: 1

23: [enemy(severus_snape, _EX4)]
min depth: 3
Nodes: 51
Answers: 24

24: [enemy(P, Z)]
min depth: 3
Nodes: 307
Answers: 270

25: [magical_pet(X, P), horcrux(P), member_of(X, death_eaters)]
min depth: 5
Nodes: 18
Answers: 4

26: [enemy(albus_dumbledore, narcissa_malfoy)]
min depth: 3
Nodes: 9
Answers: 3

27: [magical(P), magical_creature(P), creature(P)]
min depth: 1
Nodes: 10
Answers: 9

28: [descendant(X, P), betrays(percys_weasley, P), ancestor(P, X), parent(P, Y), ancestor(P, Y)]
min depth: 5
Nodes: 1594
Answers: 40

29: [in_gryffindor(_EX1)]
min depth: 2
Nodes: 20
Answers: 18

30: [enemy(X, parvati_patil)]
min depth: 3
Nodes: 24
Answers: 7

31: [subject(P)]
min depth: 1
Nodes: 12
Answers: 11

32: [witch(_EX0)]
min depth: 3

Nodes: 77
Answers: 7

33: [hero(ron_weasley)]
min depth: 2
Nodes: 3
Answers: 1

34: [hates(Y, _EX2)]
min depth: 4
Nodes: 98
Answers: 70

35: [magical_creature(P), magical(P), creature(P)]
min depth: 3
Nodes: 20
Answers: 9

36: [hates(peter_pettigrew, _EX4)]
min depth: 4
Nodes: 26
Answers: 10

37: [betrays(P, remus_lupin), enemy(P, X), enemy(narcissa_malfoy, X), enemy
(P, fred_weasley)]
min depth: 1
Nodes: 2
Answers: 1

38: [witch(narcissa_malfoy)]
min depth: 3
Nodes: 7
Answers: 1

39: [in_ravenclaw(_EX0)]
min depth: 2
Nodes: 5
Answers: 3

40: [hates(Y, _EX0)]
min depth: 4
Nodes: 98
Answers: 70

41: [enemy(_EX0, draco_malfoy)]
min depth: 3
Nodes: 36
Answers: 30

42: [enemy(_EX1, hermione_granger)]
min depth: 1
Nodes: 43
Answers: 17

43: [hates(narcissa_malfoy, _EX1)]
min depth: 4

Nodes: 28
Answers: 10

44: [hates(_EX0, _EX2)]
min depth: 4
Nodes: 94
Answers: 70

45: [house(P, nymphadora_tonks), house(P, cedric_diggory), house(P, newt_sca
mander)]
min depth: 1
Nodes: 2
Answers: 1

46: [magical_creature(Z)]
min depth: 3
Nodes: 22
Answers: 9

47: [descendant(P, arthur_weasley), parent(hermione_granger, P)]
Restart...
min depth: 3
Nodes: 1658
Answers: 14

48: [in_hufflepuff(P), female(P)]
min depth: 2
Nodes: 6
Answers: 4

49: [parent(P, harry_potter), descendant(X, P), descendant(X, harry_potter),
descendant(X, ginny_weasley), ancestor(ginny_weasley, X)]
min depth: 1
Nodes: 3
Answers: 2

50: [in_slytherin(_EX1)]
min depth: 2
Nodes: 13
Answers: 11

51: [enemy(P, X), hates(P, argus_filch), enemy(dolores_umbridge, X)]
min depth: 1
Nodes: 301
Answers: 273

52: [evil(_EX3)]
min depth: 3
Nodes: 10
Answers: 7

53: [in_slytherin(P), house(X, P), house(X, Y), hates(Y, Z), wizard(Z)]
min depth: 2
Nodes: 13
Answers: 11

54: [wizard(_EX2)]
min depth: 3
Nodes: 85
Answers: 17

55: [descendant(X, P), enemy(P, severus_snape), ancestor(P, bill_weasley), descendant(X, lily_potter), betrays(percys_weasley, P)]
min depth: 3
Nodes: 351
Answers: 47

56: [ancestor(_EX2, Z)]
Restart...
min depth: 2
Nodes: 14840
Answers: 45

57: [male(P), in_slytherin(P)]
min depth: 1
Nodes: 44
Answers: 43

58: [enemy(P, draco_malfoy), enemy(P, X), male(P), hates(X, marge_dursley)]
min depth: 3
Mismatched predicates: enemy(P, X) with member_of(draco_malfoy, dumbledores_army)
Nodes: 21
Answers: 9

59: [ancestor(harry_potter, _EX5)]
Restart...
min depth: 2
Nodes: 5590
Answers: 3

60: [wizard(P), member_of(P, X), member_of(minerva_mccgonagall, X), parent(P, fred_weasley)]
min depth: 3
Nodes: 85
Answers: 17

61: [subject(P), teaches(X, P), enemy(voldemort, X)]
min depth: 1
Nodes: 12
Answers: 11

62: [deatheater(_EX2)]
min depth: 2
Nodes: 9
Answers: 7

63: [professor(P), male(P)]
min depth: 1
Nodes: 14
Answers: 13

64: [teaches(P, care_of_magical_creatures), enemy(voldemort, P), magical_pet(P, aragog)]
min depth: 1
Nodes: 2
Answers: 1

65: [descendant(_EX4, james_potter)]
Restart...
min depth: 5
Nodes: 350
Answers: 4

66: [teaches(P, X), magical_pet(P, fluffy), subject(X)]
min depth: 1
Nodes: 14
Answers: 13

67: [in_slytherin(_EX5)]
min depth: 2
Nodes: 13
Answers: 11

68: [enemy(peter_pettigrew, _EX1)]
min depth: 1
Nodes: 25
Answers: 12

69: [subject(P), teaches(X, P), enemy(X, Y), ancestor(Y, rose_weasley)]
min depth: 1
Nodes: 12
Answers: 11

70: [enemy(molly_weasley, _EX2)]
min depth: 3
Nodes: 19
Answers: 8

71: [house(P, gregory_goyle), house(P, vincent_crabbe), house(P, severus_snape)]
min depth: 1
Nodes: 2
Answers: 1

72: [descendant(Y, _EX3)]
min depth: 3
Nodes: 1388
Answers: 45

73: [in_gryffindor(P)]
min depth: 2
Nodes: 20
Answers: 18

74: [enemy(P, Z)]
min depth: 3
Nodes: 243

Answers: 214

75: [in_gryffindor(_EX2)]
min depth: 2
Nodes: 20
Answers: 18

76: [enemy(P, _EX2)]
min depth: 1
Nodes: 232
Answers: 207

77: [subject(P)]
min depth: 1
Nodes: 12
Answers: 11

78: [enemy(remus_lupin, _EX3)]
min depth: 3
Nodes: 40
Answers: 21

79: [professor(P), teaches(P, potions), male(P), house(slytherin, P)]
min depth: 1
Nodes: 14
Answers: 13

80: [magical(P), magical_creature(P), creature(P)]
min depth: 1
Nodes: 10
Answers: 9

81: [wizard(P), enemy(draco_malfoy, P)]
min depth: 3
Nodes: 124
Answers: 17

82: [ancestor(james_potter, Y)]
min depth: 2
Nodes: 935
Answers: 4

83: [ancestor(X, _EX3)]
min depth: 2
Nodes: 19871
Answers: 51

84: [in_gryffindor(P)]
min depth: 2
Nodes: 20
Answers: 18

85: [hates(_EX1, _EX0)]
min depth: 4
Nodes: 98
Answers: 70

86: [destroyed(P, Y), parent(X, P), hates(lucius_malfoy, X), horcrux(Y)]
min depth: 1
Nodes: 7
Answers: 6

87: [descendant(_EX3, Z)]
min depth: 3
Nodes: 3537
Answers: 46

88: [hates(_EX4, arabella_figg)]
min depth: 4
Nodes: 19
Answers: 7

89: [enemy(lucius_malfoy, Z)]
min depth: 3
Nodes: 25
Answers: 9

90: [enemy(lucius_malfoy, P), enemy(X, P), enemy(X, wizarding_world), enemy(severus_snape, P)]
min depth: 3
Nodes: 41
Answers: 26

91: [hates(P, _EX3)]
min depth: 4
Nodes: 98
Answers: 70

92: [in_gryffindor(fred_weasley)]
min depth: 2
Nodes: 3
Answers: 1

93: [descendant(Y, Z)]
Restart...
Restart...
Answers: 0

94: [male(P), wizard(P)]
min depth: 1
Nodes: 44
Answers: 43

95: [enemy(lucius_malfoy, P)]
min depth: 3
Nodes: 22
Answers: 13

96: [enemy(arthur_weasley, _EX5)]
min depth: 3
Nodes: 26
Answers: 7

97: [teaches(X, P), subject(P), professor(X)]
min depth: 1
Nodes: 14
Answers: 13

98: [ancestor(P, james_sirius_potter), parent(P, fred_weasley)]
min depth: 2
Nodes: 862
Answers: 10

99: [enemy(arthur_weasley, Y)]
min depth: 3
Nodes: 35
Answers: 14

100: [enemy(P, X), enemy(arthur_weasley, P), in_gryffindor(X), member_of(X, order_of_the_phoenix)]
min depth: 1
Nodes: 230
Answers: 206

Before removing duplicates: 560467 examples
After removing duplicates: 4348 examples
Replicating pos examples 1 times
pos len=2916, neg len=2916

Step #6: Train the scoring model

Train the scoring model. This step will usually take several minutes, especially if you are not using a GPU (for the gameofthrones KB, it takes about 30 minutes on CPU).

Creates a file:

- uni_mr_model.pt - Contains the weights for the learned scoring model

```
In [10]: !python nnreasoner.py --embed_type unification -s potter_model.pt
```

States from vocab: vocab
Num predicates: 46
Max arity: 3
Num constants: 110
Num variables: 10

Training unification

Embed size: 50

Reading examples from mr_train_examples.csv

Parsing examples...

Generating embeddings...

Embedding input size: 406

[REDACTED] 5831/5831 Prepping examples

Loaded data...

Using cuda device

/opt/tljh/user/envs/cse327python311/lib/python3.11/site-packages/torch/nn/modules/lazy.py:180: UserWarning: Lazy modules are a new feature under heavy development so changes to the API or functionality can happen at any moment.

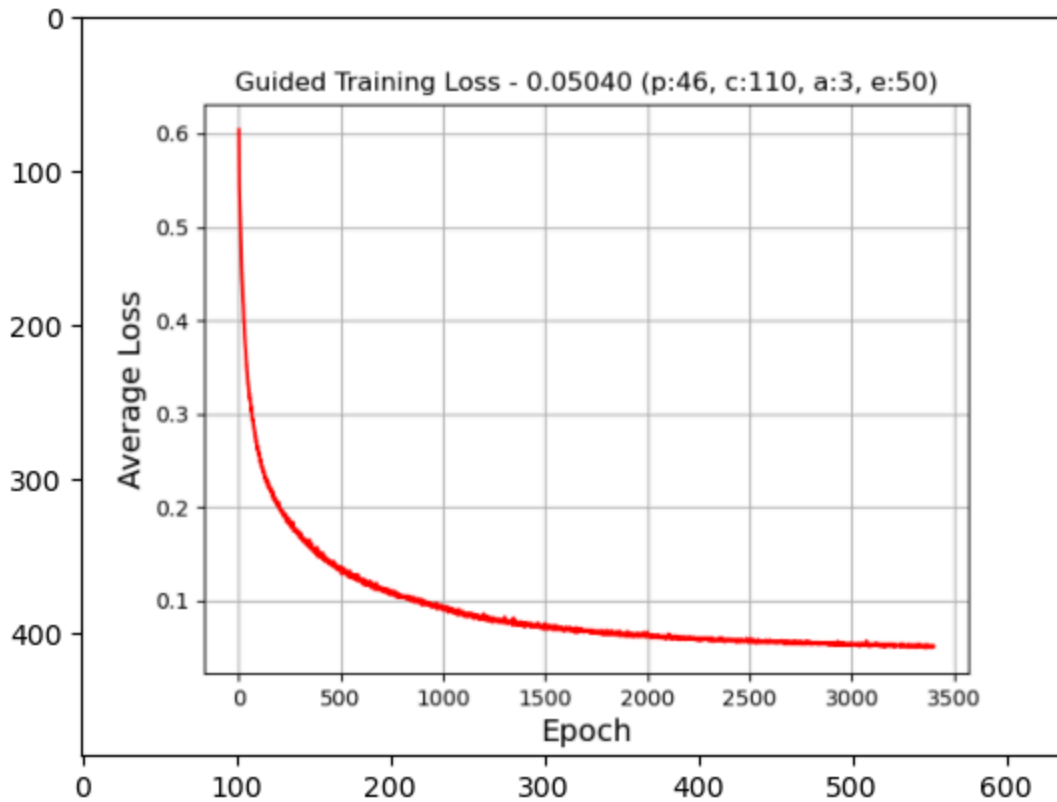
warnings.warn('Lazy modules are a new feature under heavy development ')

0	0.60424	(-)
50	0.32222	(-0.0052)
100	0.25458	(-0.0033)
150	0.22193	(-0.0024)
200	0.19660	(-0.0019)
250	0.18151	(-0.0016)
300	0.17048	(-0.0014)
350	0.15580	(-0.0012)
400	0.14738	(-0.0011)
450	0.13977	(-0.0010)
500	0.13298	(-0.0009)
550	0.12738	(-0.0008)
600	0.12208	(-0.0007)
650	0.11658	(-0.0007)
700	0.11259	(-0.0006)
750	0.10820	(-0.0006)
800	0.10394	(-0.0006)
850	0.10178	(-0.0005)
900	0.10004	(-0.0005)
950	0.09595	(-0.0005)
1000	0.09296	(-0.0005)
1050	0.08676	(-0.0004)
1100	0.08481	(-0.0004)
1150	0.08468	(-0.0004)
1200	0.08170	(-0.0004)
1250	0.07864	(-0.0004)
1300	0.08223	(-0.0004)
1350	0.07751	(-0.0003)
1400	0.07508	(-0.0003)
1450	0.07272	(-0.0003)
1500	0.07042	(-0.0003)
1550	0.07266	(-0.0003)
1600	0.06790	(-0.0003)
1650	0.07057	(-0.0003)
1700	0.06784	(-0.0003)
1750	0.06646	(-0.0003)
1800	0.06537	(-0.0002)

```
1850    0.06524 (-0.0002)
1900    0.06670 (-0.0002)
1950    0.06308 (-0.0002)
2000    0.06336 (-0.0002)
2050    0.06105 (-0.0002)
2100    0.06111 (-0.0002)
2150    0.06012 (-0.0002)
2200    0.06002 (-0.0002)
2250    0.05855 (-0.0002)
2300    0.05753 (-0.0002)
2350    0.05944 (-0.0002)
2400    0.05691 (-0.0002)
2450    0.05733 (-0.0002)
2500    0.05608 (-0.0002)
2550    0.05606 (-0.0002)
2600    0.05468 (-0.0002)
2650    0.05498 (-0.0002)
2700    0.05490 (-0.0002)
2750    0.05509 (-0.0001)
2800    0.05333 (-0.0001)
2850    0.05414 (-0.0001)
2900    0.05213 (-0.0001)
2950    0.05264 (-0.0001)
3000    0.05224 (-0.0001)
3050    0.05290 (-0.0001)
3100    0.05144 (-0.0001)
3150    0.05086 (-0.0001)
3200    0.05132 (-0.0001)
3250    0.05188 (-0.0001)
3300    0.05159 (-0.0001)
3350    0.04965 (-0.0001)
3399    0.05040 (-0.0001)
Saved training loss figure.
```

```
In [11]: img = plt.imread('guided_loss-unification46-110-3-50.png')
         plt.imshow(img)
```

```
Out[11]: <matplotlib.image.AxesImage at 0x71d630440ed0>
```



Step #7: Evaluate the scoring model -- compare std (-s) with scoring (-u)

Evaluate the scoring model by running `evaluate.py`. We compare the standard backward-chaining algorithm (-s) with the one that uses our learned scoring model (-u).

The output lists each query, and then provides a line that gives information about query execution. These are: number of nodes explored in the tree, depth of the solution, execution time in seconds, and the speed of exploring the tree (nodes per second).

The names of the files produced depend on the KB used and the options selected.

The code below produces the files:

- `std-45-106-2-50.csv` - A CSV file with query results for the standard backward-chaining algorithm
- `ming-45-106-2-50.csv` - A CSV file with query results for the learned reasoner. `ming` stands for "min scoring goal" and is the default strategy for deciding how to make use of the scores

If you use a different knowledge base, you will likely get different file names that depend on the features of that KB's vocabulary.

```
In [12]: !python evaluate.py --kb potter.txt -s
         !python evaluate.py --kb potter.txt -u --scoring_model_path potter_model.pt
```

using cuda device
STANDARD

Query 1: [destroyed(ron_weasley, P), horcrux(P), destroyed_horcrux(ron_weasley, P)]

6 :: 5 - 00:00:00 (12448 nps)

Query 2: [enemy(sirius_black, Y)]

4 :: 3 - 00:00:00 (4676 nps)

Query 3: [in_gryffindor(petter_pettigrew)]

3 :: 2 - 00:00:00 (10368 nps)

Query 4: [descendant(_EX2, P)]

4 :: 3 - 00:00:00 (8559 nps)

Query 5: [descendant(hugo_weasley, _EX3)]

4 :: 3 - 00:00:00 (11007 nps)

Query 6: [hates(bellatrix_lestrange, Y)]

5 :: 4 - 00:00:00 (13215 nps)

Query 7: [enemy(_EX2, _EX4)]

2 :: 1 - 00:00:00 (4463 nps)

Query 8: [descendant(james_sirius_potter, X)]

4 :: 3 - 00:00:00 (11918 nps)

Query 9: [enemy(_EX4, fred_weasley)]

2 :: 1 - 00:00:00 (6410 nps)

Query 10: [in_gryffindor(remus_lupin)]

3 :: 2 - 00:00:00 (10515 nps)

Query 11: [enemy(_EX3, lucius_malfoy)]

4 :: 3 - 00:00:00 (3409 nps)

Query 12: [enemy(remus_lupin, X)]

4 :: 3 - 00:00:00 (4856 nps)

Query 13: [hates(P, petunia_dursley), enemy(nymphadora_tonks, P)]

26 :: 7 - 00:00:00 (19995 nps)

Query 14: [parent(X, P), enemy(harry_potter, P), likes_voldemort(X), enemy(P, george_weasley)]

76 :: 10 - 00:00:00 (11235 nps)

Query 15: [enemy(_EX2, george_weasley)]

2 :: 1 - 00:00:00 (7062 nps)

Query 16: [creature(P)]

2 :: 1 - 00:00:00 (12504 nps)

Query 17: [in_gryffindor(_EX2)]

3 :: 2 - 00:00:00 (8113 nps)

Query 18: [male(P), enemy(P, X), enemy(lucius_malfoy, X)]

85 :: 7 - 00:00:00 (8832 nps)

Query 19: [enemy(draco_malfoy, P), enemy(bellatrix_lestrange, P)]

6 :: 4 - 00:00:00 (5158 nps)

Query 20: [subject(P)]

2 :: 1 - 00:00:00 (15609 nps)

Query 21: [in_gryffindor(_EX0)]

3 :: 2 - 00:00:00 (8033 nps)

Query 22: [female(P), house(X, P), house(X, nymphadora_tonks), teaches(P, herbology)]

23 :: 4 - 00:00:00 (5120 nps)

Query 23: [enemy(X, _EX2)]

2 :: 1 - 00:00:00 (4586 nps)

Query 24: [ancestor(arthur_weasley, Z)]

3 :: 2 - 00:00:00 (8137 nps)

Query 25: [ancestor(harry_potter, _EX0)]

3 :: 2 - 00:00:00 (8514 nps)

Query 26: [wizard(P)]

4 :: 3 - 00:00:00 (8171 nps)

Query 27: [ancestor(harry_potter, P), descendant(P, lily_potter), female(P)]

168 :: 8 - 00:00:00 (7618 nps)

Query 28: [enemy(sirius_black, _EX5)]

4 :: 3 - 00:00:00 (6081 nps)

Query 29: [enemy(_EX4, peter_pettigrew)]

4 :: 3 - 00:00:00 (7197 nps)

Query 30: [hates(Z, _EX2)]

5 :: 4 - 00:00:00 (12805 nps)

Query 31: [enemy(molly_weasley, _EX3)]

4 :: 3 - 00:00:00 (3289 nps)

Query 32: [male(P), enemy(P, X), enemy(X, Y), ancestor(Y, james_sirius_potter), parent(Y, lily_luna_potter)]

73 :: 8 - 00:00:00 (22292 nps)

Query 33: [descendant(P, Y), descendant(P, X), enemy(X, bellatrix_lestrange), parent(Y, albus_severus_potter), ancestor(Y, P)]

1165 :: 12 - 00:00:00 (5263 nps)

Query 34: [in_gryffindor(P)]

3 :: 2 - 00:00:00 (7255 nps)

Query 35: [enemy(X, seamus_finnigan)]

15 :: 3 - 00:00:00 (12060 nps)

Query 36: [hates(P, X), hates(P, myrtle_warren), hates(bellatrix_lestrange, X)]
13 :: 12 - 00:00:00 (3451 nps)

Query 37: [magical(P), creature(P), magical_creature(P)]
6 :: 5 - 00:00:00 (16018 nps)

Query 38: [witch(lily_potter)]
8 :: 3 - 00:00:00 (20119 nps)

Query 39: [hates(_EX3, petunia_dursley)]
23 :: 4 - 00:00:00 (90924 nps)

Query 40: [male(P), ancestor(P, X), descendant(X, P), parent(P, X), wizard(X)]
13 :: 10 - 00:00:00 (9859 nps)

Query 41: [house(X, P), in_hufflepuff(P), house(X, nymphadora_tonks)]
63 :: 4 - 00:00:00 (7706 nps)

Query 42: [magical_creature(P), magical(P), creature(P)]
7 :: 5 - 00:00:00 (17116 nps)

Query 43: [enemy(X, _EX5)]
2 :: 1 - 00:00:00 (4798 nps)

Query 44: [magical_creature(X)]
5 :: 3 - 00:00:00 (29309 nps)

Query 45: [descendant(_EX1, _EX3)]
4 :: 3 - 00:00:00 (8745 nps)

Query 46: [hates(_EX4, _EX5)]
5 :: 4 - 00:00:00 (9268 nps)

Query 47: [enemy(_EX5, neville_longbottom)]
2 :: 1 - 00:00:00 (6595 nps)

Query 48: [descendant(_EX5, _EX2)]
4 :: 3 - 00:00:00 (8533 nps)

Query 49: [descendant(james_sirius_potter, Z)]
4 :: 3 - 00:00:00 (12125 nps)

Query 50: [betrays(P, ron_weasley), betrays(P, charlie_weasley)]
5 :: 2 - 00:00:00 (7682 nps)

Query 51: [enemy(P, bellatrix_lestrange), betrays(peter_pettigrew, P)]
13 :: 4 - 00:00:00 (9890 nps)

Query 52: [hates(Z, P)]
5 :: 4 - 00:00:00 (15613 nps)

Query 53: [enemy(_EX2, _EX3)]
2 :: 1 - 00:00:00 (5047 nps)

Query 54: [enemy(_EX3, neville_longbottom)]

2 :: 1 - 00:00:00 (7255 nps)

Query 55: [horcrux(P), destroyed_horcrux(hermione_granger, P), destroyed(hermione_granger, P)]

18 :: 5 - 00:00:00 (23052 nps)

Query 56: [wizard(P), descendant(P, X), enemy(X, ron_weasley)]

272 :: 9 - 00:00:00 (22265 nps)

Query 57: [subject(P)]

2 :: 1 - 00:00:00 (41578 nps)

Query 58: [hates(Z, X)]

5 :: 4 - 00:00:00 (41852 nps)

Query 59: [enemy(_EX0, _EX3)]

2 :: 1 - 00:00:00 (4890 nps)

Query 60: [enemy(P, ron_weasley), male(P)]

3 :: 2 - 00:00:00 (5609 nps)

Query 61: [evil(Y)]

4 :: 3 - 00:00:00 (42058 nps)

Query 62: [hates(voldemort, P), hates(narcissa_malfoy, P)]

9 :: 8 - 00:00:00 (13356 nps)

Query 63: [enemy(_EX4, _EX2)]

2 :: 1 - 00:00:00 (3432 nps)

Query 64: [teaches(X, P), subject(P), professor(X)]

4 :: 3 - 00:00:00 (11191 nps)

Query 65: [enemy(_EX1, hermione_granger)]

2 :: 1 - 00:00:00 (6794 nps)

Query 66: [hates(_EX3, lily_potter)]

5 :: 4 - 00:00:00 (33631 nps)

Query 67: [subject(P)]

2 :: 1 - 00:00:00 (47193 nps)

Query 68: [enemy(Y, severus_snape)]

2 :: 1 - 00:00:00 (7434 nps)

Query 69: [house(P, newt_scamander), house(P, X), house(P, Y), teaches(X, herbology), male(Y)]

21 :: 5 - 00:00:00 (14857 nps)

Query 70: [teaches(P, potions), house(X, P), house(X, regulus_black), male(P)]

5 :: 4 - 00:00:00 (6090 nps)

Query 71: [descendant(X, _EX5)]

4 :: 3 - 00:00:00 (6424 nps)

Query 72: [ancestor(arthur_weasley, _EX1)]

3 :: 2 - 00:00:00 (8169 nps)

Query 73: [ancestor(james_potter, P), descendant(P, lily_potter), parent(ginny_weasley, P)]

83 :: 10 - 00:00:00 (13095 nps)

Query 74: [creature(P)]

2 :: 1 - 00:00:00 (38604 nps)

Query 75: [hates(_EX3, _EX5)]

5 :: 4 - 00:00:00 (40319 nps)

Query 76: [enemy(P, X)]

2 :: 1 - 00:00:00 (3404 nps)

Query 77: [in_slytherin(X)]

3 :: 2 - 00:00:00 (8089 nps)

Query 78: [ancestor(_EX0, Y)]

3 :: 2 - 00:00:00 (6825 nps)

Query 79: [enemy(_EX1, X)]

2 :: 1 - 00:00:00 (4969 nps)

Query 80: [ancestor(_EX0, lily_luna_potter)]

3 :: 2 - 00:00:00 (10025 nps)

Query 81: [wizard(_EX0)]

4 :: 3 - 00:00:00 (1168 nps)

Query 82: [enemy(arthur_weasley, P), enemy(X, P), enemy(X, bellatrix_lestrange), member_of(P, death_eaters), member_of(X, order_of_the_phoenix)]

12 :: 11 - 00:00:00 (4297 nps)

Query 83: [house(P, X), house(P, charlie_weasley), betrays(Y, X), enemy(harry_potter, Y), enemy(minerva_mccgonagall, Y)]

26 :: 9 - 00:00:00 (5907 nps)

Query 84: [wizard(_EX2)]

4 :: 3 - 00:00:00 (9358 nps)

Query 85: [enemy(_EX5, X)]

2 :: 1 - 00:00:00 (4806 nps)

Query 86: [destroyed_horcrux(P, tom_riddles_diary), wizard(P), enemy(narcissa_malfoy, P)]

13 :: 9 - 00:00:00 (11563 nps)

Query 87: [likes_voldemort(_EX5)]

4 :: 3 - 00:00:00 (39457 nps)

Query 88: [hates(bellatrix_lestrange, _EX0)]

5 :: 4 - 00:00:00 (32696 nps)

```

Query 89: [deatheater(_EX5)]
3 :: 2 - 00:00:00 (9993 nps)

Query 90: [descendant(_EX3, alice_longbottom)]
4 :: 3 - 00:00:00 (8032 nps)

Query 91: [likes_voldemort(_EX4)]
4 :: 3 - 00:00:00 (48008 nps)

Query 92: [in_slytherin(narcissa_malfoy)]
3 :: 2 - 00:00:00 (10461 nps)

Query 93: [pure_blood(P)]
2 :: 1 - 00:00:00 (11306 nps)

Query 94: [betrays(P, X), betrays(P, hermione_granger), betrays(percyy_weasley, X)]
8 :: 3 - 00:00:00 (8496 nps)

Query 95: [descendant(dudley_dursley, petunia_dursley)]
4 :: 3 - 00:00:00 (12277 nps)

Query 96: [in_hufflepuff(_EX0)]
3 :: 2 - 00:00:00 (8274 nps)

Query 97: [hates(P, colin_creevey), enemy(X, P), descendant(Y, X), wizard(Y)]
474 :: 13 - 00:00:00 (24526 nps)

Query 98: [subject(P)]
2 :: 1 - 00:00:00 (38884 nps)

Query 99: [teaches(P, Y), enemy(P, X), hates(X, petunia_dursley), teaches(gilderoy_lockhart, Y), pure_blood(X)]
358 :: 10 - 00:00:00 (19785 nps)

Query 100: [hates(peter_pettigrew, vernon_dursley)]
11 :: 4 - 00:00:00 (44923 nps)

std: 33.26 nodes/query
Time to run all queries: 0.3859232040000036
using cuda device
UNITY: ming
    Embedding Model: rKB_model.pth
    Scoring Model: potter_model.pt

    Control: mingoal
    Rule Eval: Max Rule
Learned Embedding: Network using cuda
/opt/tljh/user/envs/cse327python311/lib/python3.11/site-packages/torch/nn/modules/lazy.py:180: UserWarning: Lazy modules are a new feature under heavy development so changes to the API or functionality can happen at any moment.
  warnings.warn('Lazy modules are a new feature under heavy development ')
Query 1: [destroyed(ron_weasley, P), horcrux(P), destroyed_horcrux(ron_weasley, P)]

```

4 :: 3 - 00:00:00 (8 nps)

Query 2: [enemy(sirius_black, Y)]

4 :: 3 - 00:00:00 (118 nps)

Query 3: [in_gryffindor(petter_pettigrew)]

3 :: 2 - 00:00:00 (836 nps)

Query 4: [descendant(_EX2, P)]

4 :: 3 - 00:00:00 (63 nps)

Query 5: [descendant(hugo_weasley, _EX3)]

4 :: 3 - 00:00:00 (382 nps)

Query 6: [hates(bellatrix_lestrange, Y)]

5 :: 4 - 00:00:00 (267 nps)

Query 7: [enemy(_EX2, _EX4)]

2 :: 1 - 00:00:00 (34 nps)

Query 8: [descendant(james_sirius_potter, X)]

6 :: 5 - 00:00:00 (48 nps)

Query 9: [enemy(_EX4, fred_weasley)]

2 :: 1 - 00:00:00 (290 nps)

Query 10: [in_gryffindor(remus_lupin)]

3 :: 2 - 00:00:00 (877 nps)

Query 11: [enemy(_EX3, lucius_malfoy)]

4 :: 3 - 00:00:00 (110 nps)

Query 12: [enemy(remus_lupin, X)]

4 :: 3 - 00:00:00 (141 nps)

Query 13: [hates(P, petunia_dursley), enemy(nymphadora_tonks, P)]

16 :: 6 - 00:00:00 (220 nps)

Query 14: [parent(X, P), enemy(harry_potter, P), likes_voldemort(X), enemy(P, george_weasley)]

81 :: 10 - 00:00:00 (291 nps)

Query 15: [enemy(_EX2, george_weasley)]

2 :: 1 - 00:00:00 (273 nps)

Query 16: [creature(P)]

2 :: 1 - 00:00:00 (149 nps)

Query 17: [in_gryffindor(_EX2)]

3 :: 2 - 00:00:00 (111 nps)

Query 18: [male(P), enemy(P, X), enemy(lucius_malfoy, X)]

727 :: 7 - 00:00:00 (979 nps)

Query 19: [enemy(draco_malfoy, P), enemy(bellatrix_lestrange, P)]

6 :: 4 - 00:00:00 (214 nps)

Query 20: [subject(P)]
2 :: 1 – 00:00:00 (170 nps)

Query 21: [in_gryffindor(_EX0)]
3 :: 2 – 00:00:00 (21 nps)

Query 22: [female(P), house(X, P), house(X, nymphadora_tonks), teaches(P, herbology)]
5 :: 4 – 00:00:00 (56 nps)

Query 23: [enemy(X, _EX2)]
2 :: 1 – 00:00:00 (42 nps)

Query 24: [ancestor(arthur_weasley, Z)]
3 :: 2 – 00:00:00 (213 nps)

Query 25: [ancestor(harry_potter, _EX0)]
3 :: 2 – 00:00:00 (311 nps)

Query 26: [wizard(P)]
6 :: 3 – 00:00:00 (207 nps)

Query 27: [ancestor(harry_potter, P), descendant(P, lily_potter), female(P)]
9 :: 7 – 00:00:00 (51 nps)

Query 28: [enemy(sirius_black, _EX5)]
4 :: 3 – 00:00:00 (151 nps)

Query 29: [enemy(_EX4, peter_pettigrew)]
4 :: 3 – 00:00:00 (113 nps)

Query 30: [hates(Z, _EX2)]
5 :: 4 – 00:00:00 (204 nps)

Query 31: [enemy(molly_weasley, _EX3)]
4 :: 3 – 00:00:00 (144 nps)

Query 32: [male(P), enemy(P, X), enemy(X, Y), ancestor(Y, james_sirius_potter), parent(Y, lily_luna_potter)]
564498 :: 10 – 00:00:37 (14872 nps)

Query 33: [descendant(P, Y), descendant(P, X), enemy(X, bellatrix_lestrange), parent(Y, albus_severus_potter), ancestor(Y, P)]
13 :: 11 – 00:00:00 (18 nps)

Query 34: [in_gryffindor(P)]
3 :: 2 – 00:00:00 (111 nps)

Query 35: [enemy(X, seamus_finnigan)]
4 :: 3 – 00:00:00 (579 nps)

Query 36: [hates(P, X), hates(P, myrtle_warren), hates(bellatrix_lestrange, X)]
95 :: 12 – 00:00:00 (486 nps)

Query 37: [magical(P), creature(P), magical_creature(P)]

6 :: 5 - 00:00:00 (264 nps)

Query 38: [witch(lily_potter)]

8 :: 3 - 00:00:00 (761 nps)

Query 39: [hates(_EX3, petunia_dursley)]

5 :: 4 - 00:00:00 (491 nps)

Query 40: [male(P), ancestor(P, X), descendant(X, P), parent(P, X), wizard(X)]

1281 :: 8 - 00:00:00 (1932 nps)

Query 41: [house(X, P), in_hufflepuff(P), house(X, nymphadora_tonks)]

5 :: 4 - 00:00:00 (83 nps)

Query 42: [magical_creature(P), magical(P), creature(P)]

6 :: 5 - 00:00:00 (32933 nps)

Query 43: [enemy(X, _EX5)]

2 :: 1 - 00:00:00 (40 nps)

Query 44: [magical_creature(X)]

4 :: 3 - 00:00:00 (153 nps)

Query 45: [descendant(_EX1, _EX3)]

4 :: 3 - 00:00:00 (76 nps)

Query 46: [hates(_EX4, _EX5)]

5 :: 4 - 00:00:00 (210 nps)

Query 47: [enemy(_EX5, neville_longbottom)]

13 :: 1 - 00:00:00 (718 nps)

Query 48: [descendant(_EX5, _EX2)]

4 :: 3 - 00:00:00 (76 nps)

Query 49: [descendant(james_sirius_potter, Z)]

4 :: 3 - 00:00:00 (416 nps)

Query 50: [betrays(P, ron_weasley), betrays(P, charlie_weasley)]

3 :: 2 - 00:00:00 (339 nps)

Query 51: [enemy(P, bellatrix_lestrange), betrays(peter_pettigrew, P)]

8 :: 4 - 00:00:00 (150 nps)

Query 52: [hates(Z, P)]

5 :: 4 - 00:00:00 (399 nps)

Query 53: [enemy(_EX2, _EX3)]

2 :: 1 - 00:00:00 (40 nps)

Query 54: [enemy(_EX3, neville_longbottom)]

13 :: 1 - 00:00:00 (1902 nps)

Query 55: [horcrux(P), destroyed_horcrux(hermione_granger, P), destroyed(her

```

mione_granger, P)]
5 :: 4 - 00:00:00 (420 nps)

Query 56: [wizard(P), descendant(P, X), enemy(X, ron_weasley)]
265 :: 9 - 00:00:00 (951 nps)

Query 57: [subject(P)]
2 :: 1 - 00:00:00 (22740 nps)

Query 58: [hates(Z, X)]
5 :: 4 - 00:00:00 (228 nps)

Query 59: [enemy(_EX0, _EX3)]
2 :: 1 - 00:00:00 (38 nps)

Query 60: [enemy(P, ron_weasley), male(P)]
9 :: 4 - 00:00:00 (294 nps)

Query 61: [evil(Y)]
4 :: 3 - 00:00:00 (1082 nps)

Query 62: [hates(voldemort, P), hates(narcissa_malfoy, P)]
8 :: 7 - 00:00:00 (119 nps)

Query 63: [enemy(_EX4, _EX2)]
2 :: 1 - 00:00:00 (41 nps)

Query 64: [teaches(X, P), subject(P), professor(X)]
4 :: 3 - 00:00:00 (104 nps)

Query 65: [enemy(_EX1, hermione_granger)]
2 :: 1 - 00:00:00 (175 nps)

Query 66: [hates(_EX3, lily_potter)]
14 :: 4 - 00:00:00 (722 nps)

Query 67: [subject(P)]
2 :: 1 - 00:00:00 (22075 nps)

Query 68: [enemy(Y, severus_snape)]
2 :: 1 - 00:00:00 (311 nps)

Query 69: [house(P, newt_scamander), house(P, X), house(P, Y), teaches(X, herbolology), male(Y)]
15 :: 5 - 00:00:00 (57 nps)

Query 70: [teaches(P, potions), house(X, P), house(X, regulus_black), male(P)]
7 :: 4 - 00:00:00 (432 nps)

Query 71: [descendant(X, _EX5)]
4 :: 3 - 00:00:00 (77 nps)

Query 72: [ancestor(arthur_weasley, _EX1)]
3 :: 2 - 00:00:00 (214 nps)

```

Query 73: [ancestor(james_potter, P), descendant(P, lily_potter), parent(ginny_weasley, P)]
791 :: 9 - 00:00:01 (720 nps)

Query 74: [creature(P)]
2 :: 1 - 00:00:00 (23999 nps)

Query 75: [hates(_EX3, _EX5)]
5 :: 4 - 00:00:00 (794 nps)

Query 76: [enemy(P, X)]
2 :: 1 - 00:00:00 (26696 nps)

Query 77: [in_slytherin(X)]
3 :: 2 - 00:00:00 (163 nps)

Query 78: [ancestor(_EX0, Y)]
3 :: 2 - 00:00:00 (58 nps)

Query 79: [enemy(_EX1, X)]
2 :: 1 - 00:00:00 (41 nps)

Query 80: [ancestor(_EX0, lily_luna_potter)]
3 :: 2 - 00:00:00 (377 nps)

Query 81: [wizard(_EX0)]
13 :: 3 - 00:00:00 (167 nps)

Query 82: [enemy(arthur_weasley, P), enemy(X, P), enemy(X, bellatrix_lestrange), member_of(P, death_eaters), member_of(X, order_of_the_phoenix)]
9 :: 8 - 00:00:00 (25 nps)

Query 83: [house(P, X), house(P, charlie_weasley), betrays(Y, X), enemy(harry_potter, Y), enemy(minerva_mccgonagall, Y)]
32 :: 8 - 00:00:00 (60 nps)

Query 84: [wizard(_EX2)]
11 :: 3 - 00:00:00 (133 nps)

Query 85: [enemy(_EX5, X)]
2 :: 1 - 00:00:00 (35 nps)

Query 86: [destroyed_horcrux(P, tom_riddles_diary), wizard(P), enemy(narcissa_malfoy, P)]
167 :: 9 - 00:00:00 (397 nps)

Query 87: [likes_voldemort(_EX5)]
4 :: 3 - 00:00:00 (1290 nps)

Query 88: [hates(bellatrix_lestrange, _EX0)]
5 :: 4 - 00:00:00 (211 nps)

Query 89: [deatheater(_EX5)]
3 :: 2 - 00:00:00 (1704 nps)

Query 90: [descendant(_EX3, alice_longbottom)]

4 :: 3 - 00:00:00 (479 nps)

Query 91: [likes_voldemort(_EX4)]

4 :: 3 - 00:00:00 (337 nps)

Query 92: [in_slytherin(narcissa_malfoy)]

3 :: 2 - 00:00:00 (946 nps)

Query 93: [pure_blood(P)]

2 :: 1 - 00:00:00 (92 nps)

Query 94: [betrays(P, X), betrays(P, hermione_granger), betrays(percys_weasley, X)]

4 :: 3 - 00:00:00 (48 nps)

Query 95: [descendant(dudley_dursley, petunia_dursley)]

8338 :: 3 - 00:00:00 (11878 nps)

Query 96: [in_hufflepuff(_EX0)]

3 :: 2 - 00:00:00 (367 nps)

Query 97: [hates(P, colin_creevey), enemy(X, P), descendant(Y, X), wizard(Y)]

6374 :: 13 - 00:00:01 (4372 nps)

Query 98: [subject(P)]

2 :: 1 - 00:00:00 (18095 nps)

Query 99: [teaches(P, Y), enemy(P, X), hates(X, petunia_dursley), teaches(gilderoy_lockhart, Y), pure_blood(X)]

222 :: 10 - 00:00:00 (725 nps)

Query 100: [hates(peter_pettigrew, vernon_dursley)]

5 :: 4 - 00:00:00 (351 nps)

ming: 5832.96 nodes/query

Time to run all queries: 49.02439840899998

The results suggest only minor improvements for the gameofthrones KB: on average 3.75 vs. 4.39 node per query when I ran it. You may get different results due to the randomization of training and random selection of test queries. Larger, more complex KBs can see much larger improvements, sometimes an order of magnitude or more.

The code below shows the first 10 lines of each query result file. A given line has:

- query sequence number
- configuration name
- number of nodes explored
- depth of solution (0 if no solution found)
- did the system find a solution (true/false)
- time in seconds

You can import these files into a spreadsheet in order to create plots or summary statistics.

```
In [13]: !head std-46-110-3-50.csv
!head ming-46-110-3-50.csv

query,std reasoner,std nodes explored,std min depth,success,time
1,std,6,5,True,0.00048197499999957927
2,std,4,3,True,0.000855352000000309
3,std,3,2,True,0.00028933999999924964
4,std,4,3,True,0.0004673280000000446
5,std,4,3,True,0.0003633839999999111
6,std,5,4,True,0.00037833699999989534
7,std,2,1,True,0.0004480410000002877
8,std,4,3,True,0.00033560099999974113
9,std,2,1,True,0.00031201100000011195
query,ming reasoner,ming nodes explored,ming min depth,success,time
1,ming,4,3,True,0.4758266000000013
2,ming,4,3,True,0.033635996000001
3,ming,3,2,True,0.0035882779999987235
4,ming,4,3,True,0.06324634099999926
5,ming,4,3,True,0.010456055000000575
6,ming,5,4,True,0.018667372999999543
7,ming,2,1,True,0.05819118199999984
8,ming,6,5,True,0.12488582199999954
9,ming,2,1,True,0.006888962999999748
```

Potter KB with Modified Embedding Size (embed size = 25)

Step #1: Imports

```
In [3]: import matplotlib.pyplot as plt

%pwd
%ls
```

```

atomgenerator.py
autoencoder.py
basictypes.py
chainbased.py
'CSE 327 P4 200-7.ipynb'
'CSE 327 P4 200-8-Copy1.ipynb'
embedmodel.py
evaluate.py
gameofthrones.txt
gen_queries.py
helpers/
kbencoder.py
kbparser.py
knowledgebase.py
mr_back_reasoner.py
mr_train_examples.csv
nnreasoner.py
nnunifier.py

potter/
potter_embed25_model.pth
potter-P4.ipynb
potter.txt
__pycache__/
reasoner.py
size200/
termwalk.py
test_queries.txt
training_and_validation_loss.png
training_loss-10-100-4-50.png
training_loss-10-97-5-25.png
training_loss-10-97-5-50.png
training_loss-45-106-2-50.png
train_queries.txt
triplets.csv
vocab.pkl
vocab.py

```

Step #2: Train the neural net

Train a neural net to convert first-order logic atoms into meaningful numeric vectors. We call these vectors "embeddings," and the learned model the embedding model. The code creates random atoms and then uses a mechanism known as triplet loss to place unifying atoms closer together.

Creates the files:

- rKB_model.pth - Saves the learned weights for the embedding model
- training_loss-45-106-2-50.png - A plot of the training loss after each epoch. This file's name depends on the vocabulary of your KB: the four numbers are number of predicates, number of constants, max arity of the predicates, and the embedding size used. If you changed the KB, you'll need to look up the name in your drive and update the code block that displays this file.

```
In [13]: !python kbencoder.py --kb_path potter.txt --train_unification_model --embed_
```

```
Using cuda device
```

```
Uni model embed size: 25
```

```
[████████████████████] 3500/3500 Anchors generated
```

```
[████████████████████] 70000/70000 Triplets generated
```

```
[████████-----] 18089/70000 Triplet encodings
```

```
IOPub data rate exceeded.
```

```
The Jupyter server will temporarily stop sending output
to the client in order to avoid crashing it.
```

```
To change this limit, set the config variable
```

```
`--ServerApp.iopub_data_rate_limit`.
```

```
Current values:
```

```
ServerApp.iopub_data_rate_limit=1000000.0 (bytes/sec)
```

```
ServerApp.rate_limit_window=3.0 (secs)
```

```
[██████████] 49373/70000 Triplet encodings
```

IOPub data rate exceeded.

The Jupyter server will temporarily stop sending output to the client in order to avoid crashing it.

To change this limit, set the config variable

`--ServerApp.iopub\_data\_rate\_limit`.

Current values:

ServerApp.iopub_data_rate_limit=1000000.0 (bytes/sec)

ServerApp.rate_limit_window=3.0 (secs)

```
[██████████] 70000/70000 Triplet encodings
```

Early stopping triggered-----] 31.3% Epoch | [Epoch 157/500, Training Loss: 0.1041, Validation Loss: 0.4086]

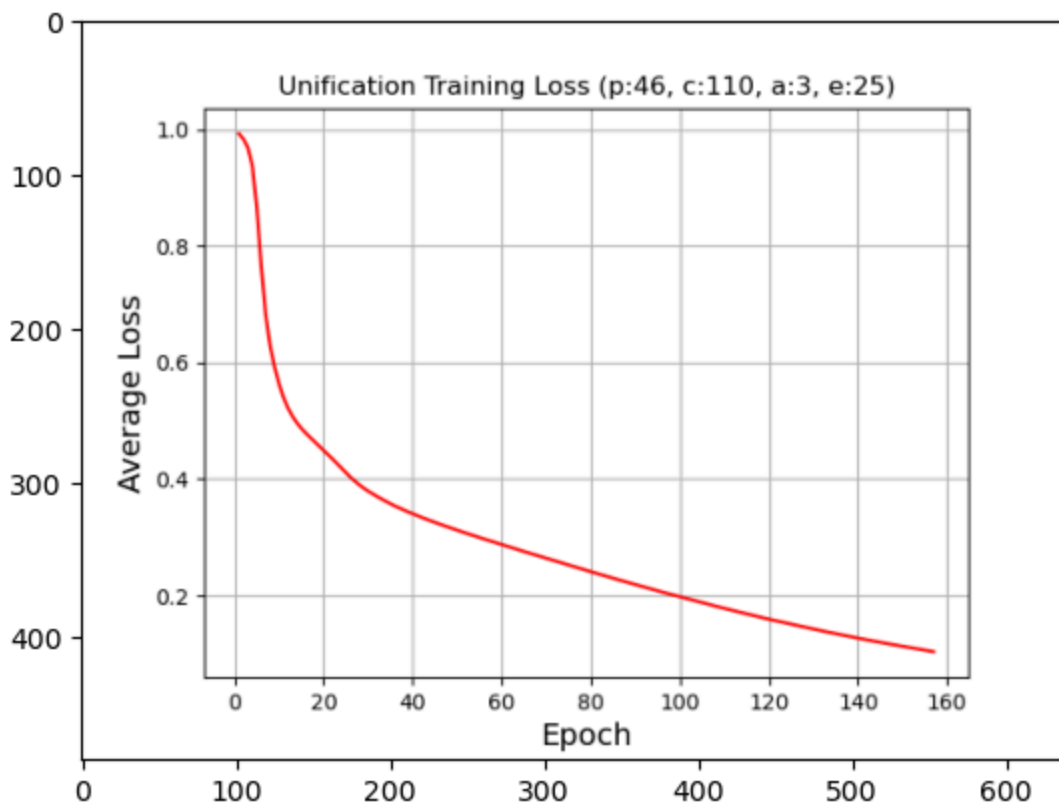
Saved model.

Saved plot.

unification model generated

```
In [15]: img = plt.imread('training_loss-46-110-3-25.png')
         plt.imshow(img)
```

```
Out[15]: <matplotlib.image.AxesImage at 0x707e63c9b110>
```



Step #3: Train the scoring model

Train the scoring model. This step will usually take several minutes, especially if you are not using a GPU (for the gameofthrones KB, it takes about 30 minutes on CPU).

Creates a file:

- .pt - Contains the weights for the learned scoring model

```
In [16]: !python nnreasoner.py --embed_model_path potter_embed25_model.pth --embed_ty
```

States from vocab: vocab
Num predicates: 46
Max arity: 3
Num constants: 110
Num variables: 10

Training unification

Embed size: 25

Reading examples from mr_train_examples.csv

Parsing examples...

Generating embeddings...

Embedding input size: 406

[██████████] 5831/5831 Prepping examples

Loaded data...

Using cuda device

/opt/tljh/user/envs/cse327python311/lib/python3.11/site-packages/torch/nn/modules/lazy.py:180: UserWarning: Lazy modules are a new feature under heavy development so changes to the API or functionality can happen at any moment.

warnings.warn('Lazy modules are a new feature under heavy development ')

0	0.62070	(-)
50	0.41744	(-0.0034)
100	0.36995	(-0.0022)
150	0.30778	(-0.0018)
200	0.26852	(-0.0016)
250	0.24985	(-0.0013)
300	0.23299	(-0.0011)
350	0.22508	(-0.0010)
400	0.21774	(-0.0009)
450	0.21446	(-0.0008)
500	0.21118	(-0.0007)
550	0.20247	(-0.0007)
600	0.19790	(-0.0006)
650	0.19090	(-0.0006)
700	0.18680	(-0.0005)
750	0.19470	(-0.0005)
800	0.18928	(-0.0005)
850	0.18834	(-0.0004)
900	0.18406	(-0.0004)
950	0.17715	(-0.0004)
1000	0.17946	(-0.0004)
1050	0.17280	(-0.0004)
1100	0.17271	(-0.0003)
1150	0.17255	(-0.0003)
1200	0.16467	(-0.0003)
1250	0.17131	(-0.0003)
1300	0.16483	(-0.0003)
1350	0.16657	(-0.0003)
1400	0.16254	(-0.0003)
1450	0.16223	(-0.0003)
1500	0.16034	(-0.0002)
1550	0.16281	(-0.0002)
1600	0.15782	(-0.0002)
1650	0.15724	(-0.0002)
1700	0.15742	(-0.0002)
1750	0.15595	(-0.0002)
1800	0.15259	(-0.0002)

```

1850    0.15635 (-0.0002)
1900    0.15149 (-0.0002)
1950    0.14758 (-0.0002)
2000    0.14801 (-0.0002)
2050    0.14629 (-0.0002)
2100    0.14485 (-0.0002)
2150    0.15060 (-0.0002)
2200    0.14856 (-0.0002)
2250    0.14181 (-0.0002)
2300    0.14339 (-0.0001)
2350    0.14241 (-0.0001)
2400    0.14009 (-0.0001)
2450    0.14352 (-0.0001)
2500    0.13904 (-0.0001)
2550    0.13879 (-0.0001)
2600    0.13935 (-0.0001)
2650    0.13857 (-0.0001)
2700    0.13438 (-0.0001)
2750    0.13289 (-0.0001)
2800    0.13229 (-0.0001)
2849    0.13495 (-0.0001)
Saved training loss figure.

```

```

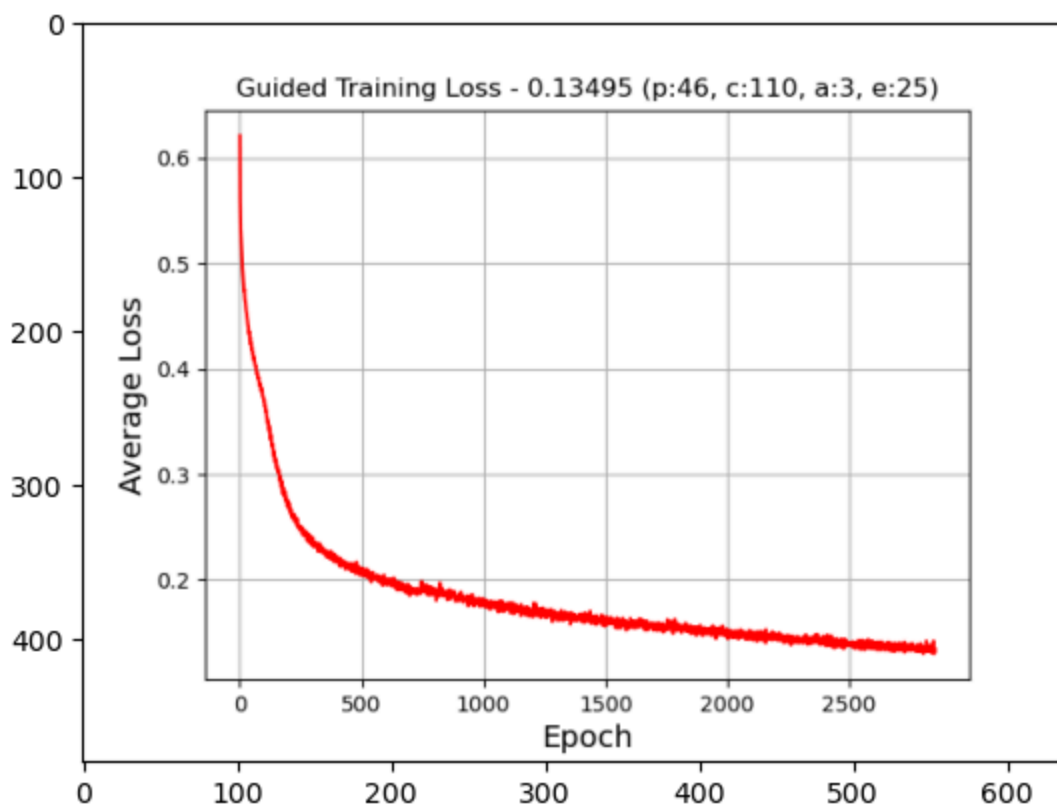
In [17]: img = plt.imread('guided_loss-unification46-110-3-25.png')
         plt.imshow(img)

```

```

Out[17]: <matplotlib.image.AxesImage at 0x707e63c5b110>

```



Step #4: Evaluate the scoring model -- compare std (-s) with scoring (-u)

Evaluate the scoring model by running `evaluate.py`. We compare the standard backward-chaining algorithm (`-s`) with the one that uses our learned scoring model (`-u`).

The output lists each query, and then provides a line that gives information about query execution. These are: number of nodes explored in the tree, depth of the solution, execution time in seconds, and the speed of exploring the tree (nodes per second).

The names of the files produced depend on the KB used and the options selected. The code below produces the files:

- `std-45-106-2-50.csv` - A CSV file with query results for the standard backward-chaining algorithm
- `ming-45-106-2-50.csv` - A CSV file with query results for the learned reasoner. `ming` stands for "min scoring goal" and is the default strategy for deciding how to make use of the scores

If you use a different knowledge base, you will likely get different file names that depend on the features of that KB's vocabulary.

```
In [19]: #!/python evaluate.py --kb potter.txt -s --embed_model_path potter_embed25_moc  
!python evaluate.py --kb potter.txt -u --embed_model_path potter_embed25_moc
```

```
using cuda device
UNITY: ming
    Embedding Model: potter_embed25_model.pth
    Scoring Model: mr_potter_embed25_model.pt

    Control: mingoal
    Rule Eval: Max Rule
Learned Embedding: Network using cuda
/opt/tljh/user/envs/cse327python311/lib/python3.11/site-packages/torch/nn/modules/lazy.py:180: UserWarning: Lazy modules are a new feature under heavy development so changes to the API or functionality can happen at any moment.
  warnings.warn('Lazy modules are a new feature under heavy development ')
Entered eval_config
Number of queries passed in: 100
Query 1: [destroyed(ron_weasley, P), horcrux(P), destroyed_horcrux(ron_weasley, P)]
10 :: 3 - 00:00:00 (18 nps)

Query 2: [enemy(sirius_black, Y)]
18 :: 3 - 00:00:00 (313 nps)

Query 3: [in_gryffindor(petter_pettigrew)]
3 :: 2 - 00:00:00 (568 nps)

Query 4: [descendant(_EX2, P)]
4 :: 3 - 00:00:00 (47 nps)

Query 5: [descendant(hugo_weasley, _EX3)]
6 :: 5 - 00:00:00 (34 nps)

Query 6: [hates(bellatrix_lestrange, Y)]
5 :: 4 - 00:00:00 (197 nps)

Query 7: [enemy(_EX2, _EX4)]
2 :: 1 - 00:00:00 (24 nps)

Query 8: [descendant(james_sirius_potter, X)]
4 :: 3 - 00:00:00 (303 nps)

Query 9: [enemy(_EX4, fred_weasley)]
2 :: 1 - 00:00:00 (201 nps)

Query 10: [in_gryffindor(remus_lupin)]
3 :: 2 - 00:00:00 (584 nps)

Query 11: [enemy(_EX3, lucius_malfoy)]
4 :: 3 - 00:00:00 (107 nps)

Query 12: [enemy(remus_lupin, X)]
18 :: 3 - 00:00:00 (287 nps)

Query 13: [hates(P, petunia_dursley), enemy(nymphadora_tonks, P)]
11 :: 7 - 00:00:00 (122 nps)

Query 14: [parent(X, P), enemy(harry_potter, P), likes_voldemort(X), enemy(P, george_weasley)]
```

10 :: 9 – 00:00:00 (15 nps)

Query 15: [enemy(_EX2, george_weasley)]
4 :: 3 – 00:00:00 (141 nps)

Query 16: [creature(P)]
2 :: 1 – 00:00:00 (111 nps)

Query 17: [in_gryffindor(_EX2)]
3 :: 2 – 00:00:00 (65 nps)

Query 18: [male(P), enemy(P, X), enemy(lucius_malfoy, X)]
38 :: 5 – 00:00:00 (74 nps)

Query 19: [enemy(draco_malfoy, P), enemy(bellatrix_lestrange, P)]
5 :: 4 – 00:00:00 (107 nps)

Query 20: [subject(P)]
2 :: 1 – 00:00:00 (102 nps)

Query 21: [in_gryffindor(_EX0)]
3 :: 2 – 00:00:00 (65 nps)

Query 22: [female(P), house(X, P), house(X, nymphadora_tonks), teaches(P, herbiology)]
5 :: 4 – 00:00:00 (36 nps)

Query 23: [enemy(X, _EX2)]
2 :: 1 – 00:00:00 (24 nps)

Query 24: [ancestor(arthur_weasley, Z)]
8 :: 4 – 00:00:00 (59 nps)

Query 25: [ancestor(harry_potter, _EX0)]
3 :: 2 – 00:00:00 (184 nps)

Query 26: [wizard(P)]
4 :: 3 – 00:00:00 (96 nps)

Query 27: [ancestor(harry_potter, P), descendant(P, lily_potter), female(P)]
5344 :: 6 – 00:00:06 (865 nps)

Query 28: [enemy(sirius_black, _EX5)]
18 :: 3 – 00:00:00 (453 nps)

Query 29: [enemy(_EX4, peter_pettigrew)]
4 :: 3 – 00:00:00 (88 nps)

Query 30: [hates(Z, _EX2)]
5 :: 4 – 00:00:00 (119 nps)

Query 31: [enemy(molly_weasley, _EX3)]
18 :: 3 – 00:00:00 (293 nps)

Query 32: [male(P), enemy(P, X), enemy(X, Y), ancestor(Y, james_sirius_potter), parent(Y, lily_luna_potter)]

649 :: 10 – 00:00:01 (549 nps)

Query 33: [descendant(P, Y), descendant(P, X), enemy(X, bellatrix_lestrange), parent(Y, albus_severus_potter), ancestor(Y, P)]
12 :: 11 – 00:00:00 (16 nps)

Query 34: [in_gryffindor(P)]
3 :: 2 – 00:00:00 (118 nps)

Query 35: [enemy(X, seamus_finnigan)]
15 :: 3 – 00:00:00 (1888 nps)

Query 36: [hates(P, X), hates(P, myrtle_warren), hates(bellatrix_lestrange, X)]
828 :: 12 – 00:00:00 (1747 nps)

Query 37: [magical(P), creature(P), magical_creature(P)]
5 :: 3 – 00:00:00 (88 nps)

Query 38: [witch(lily_potter)]
6 :: 3 – 00:00:00 (392 nps)

Query 39: [hates(_EX3, petunia_dursley)]
14 :: 4 – 00:00:00 (382 nps)

Query 40: [male(P), ancestor(P, X), descendant(X, P), parent(P, X), wizard(X)]
188 :: 7 – 00:00:00 (295 nps)

Query 41: [house(X, P), in_hufflepuff(P), house(X, nymphadora_tonks)]
4 :: 3 – 00:00:00 (43 nps)

Query 42: [magical_creature(P), magical(P), creature(P)]
5 :: 3 – 00:00:00 (29641 nps)

Query 43: [enemy(X, _EX5)]
2 :: 1 – 00:00:00 (23 nps)

Query 44: [magical_creature(X)]
4 :: 3 – 00:00:00 (105 nps)

Query 45: [descendant(_EX1, _EX3)]
4 :: 3 – 00:00:00 (44 nps)

Query 46: [hates(_EX4, _EX5)]
5 :: 4 – 00:00:00 (135 nps)

Query 47: [enemy(_EX5, neville_longbottom)]
2 :: 1 – 00:00:00 (155 nps)

Query 48: [descendant(_EX5, _EX2)]
4 :: 3 – 00:00:00 (43 nps)

Query 49: [descendant(james_sirius_potter, Z)]
6 :: 5 – 00:00:00 (31 nps)

Query 50: [betrays(P, ron_weasley), betrays(P, charlie_weasley)]
3 :: 2 - 00:00:00 (229 nps)

Query 51: [enemy(P, bellatrix_lestrange), betrays(peter_pettigrew, P)]
5 :: 4 - 00:00:00 (204 nps)

Query 52: [hates(Z, P)]
5 :: 4 - 00:00:00 (231 nps)

Query 53: [enemy(_EX2, _EX3)]
2 :: 1 - 00:00:00 (25 nps)

Query 54: [enemy(_EX3, neville_longbottom)]
2 :: 1 - 00:00:00 (191 nps)

Query 55: [horcrux(P), destroyed_horcrux(hermione_granger, P), destroyed(hermione_granger, P)]
8 :: 3 - 00:00:00 (659 nps)

Query 56: [wizard(P), descendant(P, X), enemy(X, ron_weasley)]
317 :: 9 - 00:00:00 (753 nps)

Query 57: [subject(P)]
2 :: 1 - 00:00:00 (26082 nps)

Query 58: [hates(Z, X)]
5 :: 4 - 00:00:00 (248 nps)

Query 59: [enemy(_EX0, _EX3)]
2 :: 1 - 00:00:00 (22 nps)

Query 60: [enemy(P, ron_weasley), male(P)]
7 :: 2 - 00:00:00 (143 nps)

Query 61: [evil(Y)]
4 :: 3 - 00:00:00 (180 nps)

Query 62: [hates(voldemort, P), hates(narcissa_malfoy, P)]
12 :: 8 - 00:00:00 (138 nps)

Query 63: [enemy(_EX4, _EX2)]
2 :: 1 - 00:00:00 (22 nps)

Query 64: [teaches(X, P), subject(P), professor(X)]
18 :: 3 - 00:00:00 (222 nps)

Query 65: [enemy(_EX1, hermione_granger)]
2 :: 1 - 00:00:00 (92 nps)

Query 66: [hates(_EX3, lily_potter)]
23 :: 4 - 00:00:00 (1498 nps)

Query 67: [subject(P)]
2 :: 1 - 00:00:00 (27600 nps)

Query 68: [enemy(Y, severus_snape)]

2 :: 1 - 00:00:00 (177 nps)

Query 69: [house(P, newt_scamander), house(P, X), house(P, Y), teaches(X, herbology), male(Y)]

22 :: 5 - 00:00:00 (54 nps)

Query 70: [teaches(P, potions), house(X, P), house(X, regulus_black), male(P)]

9 :: 4 - 00:00:00 (181 nps)

Query 71: [descendant(X, _EX5)]

4 :: 3 - 00:00:00 (45 nps)

Query 72: [ancestor(arthur_weasley, _EX1)]

3 :: 2 - 00:00:00 (121 nps)

Query 73: [ancestor(james_potter, P), descendant(P, lily_potter), parent(ginny_weasley, P)]

799 :: 10 - 00:00:00 (2258 nps)

Query 74: [creature(P)]

2 :: 1 - 00:00:00 (25240 nps)

Query 75: [hates(_EX3, _EX5)]

5 :: 4 - 00:00:00 (842 nps)

Query 76: [enemy(P, X)]

2 :: 1 - 00:00:00 (25297 nps)

Query 77: [in_slytherin(X)]

3 :: 2 - 00:00:00 (177 nps)

Query 78: [ancestor(_EX0, Y)]

3 :: 2 - 00:00:00 (63 nps)

Query 79: [enemy(_EX1, X)]

2 :: 1 - 00:00:00 (28 nps)

Query 80: [ancestor(_EX0, lily_luna_potter)]

3 :: 2 - 00:00:00 (233 nps)

Query 81: [wizard(_EX0)]

4 :: 3 - 00:00:00 (42 nps)

Query 82: [enemy(arthur_weasley, P), enemy(X, P), enemy(X, bellatrix_lestrange), member_of(P, death_eaters), member_of(X, order_of_the_phoenix)]

25 :: 8 - 00:00:00 (64 nps)

Query 83: [house(P, X), house(P, charlie_weasley), betrays(Y, X), enemy(harry_potter, Y), enemy(minerva_mccgonagall, Y)]

24 :: 8 - 00:00:00 (56 nps)

Query 84: [wizard(_EX2)]

4 :: 3 - 00:00:00 (46 nps)

Query 85: [enemy(_EX5, X)]

```

2 :: 1 - 00:00:00 (28 nps)

Query 86: [destroyed_horcrux(P, tom_riddles_diary), wizard(P), enemy(narcissa_malfoy, P)]
186 :: 9 - 00:00:00 (660 nps)

Query 87: [likes_voldemort(_EX5)]
4 :: 3 - 00:00:00 (349 nps)

Query 88: [hates(bellatrix_lestrange, _EX0)]
5 :: 4 - 00:00:00 (396 nps)

Query 89: [deatheater(_EX5)]
3 :: 2 - 00:00:00 (1489 nps)

Query 90: [descendant(_EX3, alice_longbottom)]
645673 :: 3 - 00:00:24 (26374 nps)

Query 91: [likes_voldemort(_EX4)]
4 :: 3 - 00:00:00 (2318 nps)

Query 92: [in_slytherin(narcissa_malfoy)]
3 :: 2 - 00:00:00 (846 nps)

Query 93: [pure_blood(P)]
2 :: 1 - 00:00:00 (104 nps)

Query 94: [betrays(P, X), betrays(P, hermione_granger), betrays(percyy_weasley, X)]
5 :: 3 - 00:00:00 (86 nps)

Query 95: [descendant(dudley_dursley, petunia_dursley)]
4 :: 3 - 00:00:00 (696 nps)

Query 96: [in_hufflepuff(_EX0)]
3 :: 2 - 00:00:00 (475 nps)

Query 97: [hates(P, colin_creevey), enemy(X, P), descendant(Y, X), wizard(Y)]
5348 :: 13 - 00:00:01 (3686 nps)

Query 98: [subject(P)]
2 :: 1 - 00:00:00 (20207 nps)

Query 99: [teaches(P, Y), enemy(P, X), hates(X, petunia_dursley), teaches(gilderoy_lockhart, Y), pure_blood(X)]
471 :: 10 - 00:00:00 (992 nps)

Query 100: [hates(peter_pettigrew, vernon_dursley)]
8 :: 4 - 00:00:00 (628 nps)

ming: 6603.84 nodes/query
Time to run all queries: 43.158802799

```

```

In [20]: !head std-46-110-3-50.csv
         !head ming-46-110-3-25.csv

```

```
query,std reasoner,std nodes explored,std min depth,success,time
1,std,6,5,True,0.0004671110000007417
2,std,4,3,True,0.000817774000000604
3,std,3,2,True,0.0002816600000006275
4,std,4,3,True,0.0004513309999998327
5,std,4,3,True,0.00035616899999979523
6,std,5,4,True,0.0003641789999999645
7,std,2,1,True,0.00043512000000056616
8,std,4,3,True,0.0003254919999999828
9,std,2,1,True,0.0002981450000003605
query,ming reasoner,ming nodes explored,ming min depth,success,time
1,ming,10,3,True,0.5301936640000005
2,ming,18,3,True,0.05737685899999967
3,ming,3,2,True,0.005274327000000412
4,ming,4,3,True,0.08509362500000073
5,ming,6,5,True,0.17237762799999956
6,ming,5,4,True,0.025274366999999742
7,ming,2,1,True,0.08079707599999963
8,ming,4,3,True,0.013159890000000729
9,ming,2,1,True,0.009904683000000247
```

In []:

In []:

Potter KB with Modified nnreasoner

Step #1: Imports

```
In [3]: import matplotlib.pyplot as plt

%pwd
%ls
```

```

atomgenerator.py
autoencoder.py
basictypes.py
chainbased.py
'CSE 327 P4 200-7.ipynb'
'CSE 327 P4 200-8-Copy1.ipynb'
embedmodel.py
evaluate.py
gameofthrones.txt
gen_queries.py
helpers/
kbencoder.py
kbparser.py
knowledgebase.py
mr_back_reasoner.py
mr_train_examples.csv
nnreasoner.py
nnunifier.py

potter/
potter_embed25_model.pth
potter-P4.ipynb
potter.txt
__pycache__/
reasoner.py
size200/
termwalk.py
test_queries.txt
training_and_validation_loss.png
training_loss-10-100-4-50.png
training_loss-10-97-5-25.png
training_loss-10-97-5-50.png
training_loss-45-106-2-50.png
train_queries.txt
triplets.csv
vocab.pkl
vocab.py

```

Step #2: Train the neural net

Train a neural net to convert first-order logic atoms into meaningful numeric vectors. We call these vectors "embeddings," and the learned model the embedding model. The code creates random atoms and then uses a mechanism known as triplet loss to place unifying atoms closer together.

Creates the files:

- rKB_model.pth - Saves the learned weights for the embedding model
- training_loss-45-106-2-50.png - A plot of the training loss after each epoch. This file's name depends on the vocabulary of your KB: the four numbers are number of predicates, number of constants, max arity of the predicates, and the embedding size used. If you changed the KB, you'll need to look up the name in your drive and update the code block that displays this file.

```
In [21]: !python kbencoder.py --kb_path potter.txt --train_unification_model --embed_
```

```
Using cuda device
```

```
Uni model embed size: 50
```

```
[████████████████████] 3500/3500 Anchors generated
```

```
[████████████████████] 70000/70000 Triplets generated
```

```
[████████-----] 18156/70000 Triplet encodings
```

```
IOPub data rate exceeded.
```

```
The Jupyter server will temporarily stop sending output  
to the client in order to avoid crashing it.
```

```
To change this limit, set the config variable
```

```
`--ServerApp.iopub_data_rate_limit`.
```

```
Current values:
```

```
ServerApp.iopub_data_rate_limit=1000000.0 (bytes/sec)
```

```
ServerApp.rate_limit_window=3.0 (secs)
```

```
[██████████] 48880/70000 Triplet encodings
```

IOPub data rate exceeded.

The Jupyter server will temporarily stop sending output to the client in order to avoid crashing it.

To change this limit, set the config variable

`--ServerApp.iopub\_data\_rate\_limit`.

Current values:

ServerApp.iopub_data_rate_limit=1000000.0 (bytes/sec)

ServerApp.rate_limit_window=3.0 (secs)

```
[██████████] 70000/70000 Triplet encodings
```

Early stopping triggered-----] 41.5% Epoch | [Epoch 208/500, Training Loss: 0.0259, Validation Loss: 0.3404]

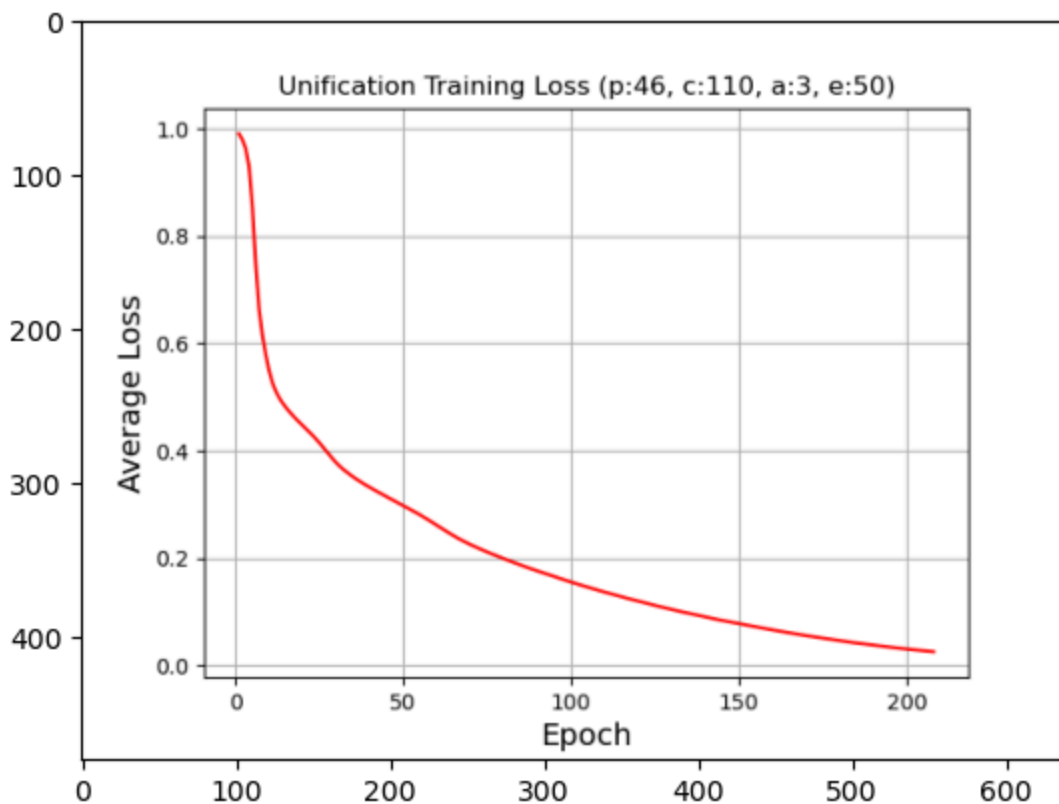
Saved model.

Saved plot.

unification model generated

```
In [22]: img = plt.imread('training_loss-46-110-3-50.png')
         plt.imshow(img)
```

```
Out[22]: <matplotlib.image.AxesImage at 0x707e62b45290>
```



Step #3: Train the scoring model

Train the scoring model. This step will usually take several minutes, especially if you are not using a GPU (for the gameofthrones KB, it takes about 30 minutes on CPU).

Creates a file:

- .pt - Contains the weights for the learned scoring model

In [24]: `!python nnreasoner-mod.py --embed_model_path potter_modnn_model.pth --embed_`

States from vocab: vocab
Num predicates: 46
Max arity: 3
Num constants: 110
Num variables: 10

Training unification

Embed size: 50

Reading examples from mr_train_examples.csv

Parsing examples...

Generating embeddings...

Embedding input size: 406

[██████████] 5831/5831 Prepping examples

Loaded data...

Using cuda device

/opt/tljh/user/envs/cse327python311/lib/python3.11/site-packages/torch/nn/modules/lazy.py:180: UserWarning: Lazy modules are a new feature under heavy development so changes to the API or functionality can happen at any moment.

warnings.warn('Lazy modules are a new feature under heavy development ')

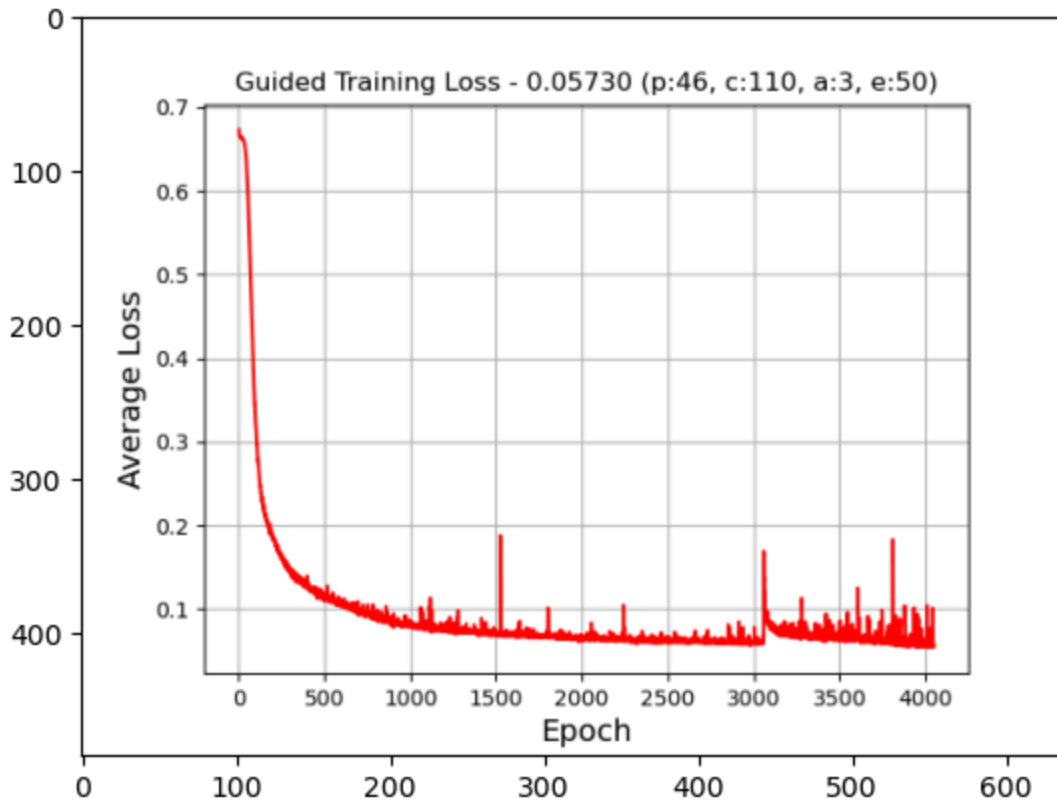
0	0.67306	(-)
50	0.60631	(-0.0013)
100	0.30989	(-0.0035)
150	0.21419	(-0.0029)
200	0.18558	(-0.0023)
250	0.16159	(-0.0019)
300	0.14317	(-0.0017)
350	0.13211	(-0.0015)
400	0.12938	(-0.0013)
450	0.12215	(-0.0012)
500	0.11425	(-0.0011)
550	0.10932	(-0.0010)
600	0.10712	(-0.0009)
650	0.10019	(-0.0008)
700	0.09477	(-0.0008)
750	0.10525	(-0.0007)
800	0.09084	(-0.0007)
850	0.08708	(-0.0006)
900	0.07807	(-0.0006)
950	0.07974	(-0.0006)
1000	0.08217	(-0.0005)
1050	0.07981	(-0.0005)
1100	0.07984	(-0.0005)
1150	0.07747	(-0.0005)
1200	0.07393	(-0.0004)
1250	0.07467	(-0.0004)
1300	0.07369	(-0.0004)
1350	0.07307	(-0.0004)
1400	0.07077	(-0.0004)
1450	0.07165	(-0.0004)
1500	0.06934	(-0.0004)
1550	0.06765	(-0.0003)
1600	0.07085	(-0.0003)
1650	0.07355	(-0.0003)
1700	0.06882	(-0.0003)
1750	0.06909	(-0.0003)
1800	0.08758	(-0.0003)

1850	0.06752	(-0.0003)
1900	0.06454	(-0.0003)
1950	0.06584	(-0.0003)
2000	0.06949	(-0.0003)
2050	0.06810	(-0.0002)
2100	0.06385	(-0.0002)
2150	0.06387	(-0.0002)
2200	0.06329	(-0.0002)
2250	0.06264	(-0.0002)
2300	0.06402	(-0.0002)
2350	0.07371	(-0.0002)
2400	0.06307	(-0.0002)
2450	0.06140	(-0.0002)
2500	0.06681	(-0.0002)
2550	0.06295	(-0.0002)
2600	0.06376	(-0.0002)
2650	0.06014	(-0.0002)
2700	0.05944	(-0.0002)
2750	0.06179	(-0.0002)
2800	0.06262	(-0.0002)
2850	0.06064	(-0.0002)
2900	0.06133	(-0.0002)
2950	0.05887	(-0.0002)
3000	0.05821	(-0.0002)
3050	0.05964	(-0.0001)
3100	0.08234	(-0.0001)
3150	0.07068	(-0.0001)
3200	0.06829	(-0.0001)
3250	0.07489	(-0.0001)
3300	0.06723	(-0.0001)
3350	0.07270	(-0.0001)
3400	0.06516	(-0.0001)
3450	0.06578	(-0.0001)
3500	0.06379	(-0.0001)
3550	0.06432	(-0.0001)
3600	0.07054	(-0.0001)
3650	0.06192	(-0.0001)
3700	0.05909	(-0.0001)
3750	0.07924	(-0.0001)
3800	0.05773	(-0.0001)
3850	0.06322	(-0.0001)
3900	0.06477	(-0.0001)
3950	0.05737	(-0.0001)
4000	0.05518	(-0.0001)
4049	0.05730	(-0.0001)

Saved training loss figure.

```
In [25]: img = plt.imread('guided_loss-unification46-110-3-50.png')
         plt.imshow(img)
```

```
Out[25]: <matplotlib.image.AxesImage at 0x707e2f799590>
```



Step #4: Evaluate the scoring model -- compare std (-s) with scoring (-u)

Evaluate the scoring model by running `evaluate.py`. We compare the standard backward-chaining algorithm (-s) with the one that uses our learned scoring model (-u).

The output lists each query, and then provides a line that gives information about query execution. These are: number of nodes explored in the tree, depth of the solution, execution time in seconds, and the speed of exploring the tree (nodes per second).

The names of the files produced depend on the KB used and the options selected.

The code below produces the files:

- `std-45-106-2-50.csv` - A CSV file with query results for the standard backward-chaining algorithm
- `ming-45-106-2-50.csv` - A CSV file with query results for the learned reasoner. `ming` stands for "min scoring goal" and is the default strategy for deciding how to make use of the scores

If you use a different knowledge base, you will likely get different file names that depend on the features of that KB's vocabulary.

```
In [28]: #!python evaluate.py --kb potter.txt -s --embed_model_path potter_embed25_model
!python evaluate.py --kb potter.txt -u --embed_model_path potter_modnn_model
```

```

using cuda device
UNITY: ming
    Embedding Model: potter_modnn_model.pth
    Scoring Model: mr_potter_modnn_model.pt

    Control: mingoal
    Rule Eval: Max Rule
Learned Embedding: Network using cuda
/opt/tljh/user/envs/cse327python311/lib/python3.11/site-packages/torch/nn/modules/lazy.py:180: UserWarning: Lazy modules are a new feature under heavy development so changes to the API or functionality can happen at any moment.
  warnings.warn('Lazy modules are a new feature under heavy development ')
Entered eval_config
Number of queries passed in: 100
Query 1: [destroyed(ron_weasley, P), horcrux(P), destroyed_horcrux(ron_weasley, P)]
4 :: 3 - 00:00:00 (8 nps)

Query 2: [enemy(sirius_black, Y)]
4 :: 3 - 00:00:00 (123 nps)

Query 3: [in_gryffindor(petter_pettigrew)]
3 :: 2 - 00:00:00 (692 nps)

Query 4: [descendant(_EX2, P)]
4 :: 3 - 00:00:00 (65 nps)

Query 5: [descendant(hugo_weasley, _EX3)]
103155 :: 5 - 00:00:10 (9606 nps)

Query 6: [hates(bellatrix_lestrange, Y)]
5 :: 4 - 00:00:00 (198 nps)

Query 7: [enemy(_EX2, _EX4)]
2 :: 1 - 00:00:00 (32 nps)

Query 8: [descendant(james_sirius_potter, X)]
28421 :: 5 - 00:00:02 (10111 nps)

Query 9: [enemy(_EX4, fred_weasley)]
2 :: 1 - 00:00:00 (242 nps)

Query 10: [in_gryffindor(remus_lupin)]
3 :: 2 - 00:00:00 (555 nps)

Query 11: [enemy(_EX3, lucius_malfoy)]
12 :: 3 - 00:00:00 (214 nps)

Query 12: [enemy(remus_lupin, X)]
4 :: 3 - 00:00:00 (185 nps)

Query 13: [hates(P, petunia_dursley), enemy(nymphadora_tonks, P)]
12 :: 7 - 00:00:00 (168 nps)

Query 14: [parent(X, P), enemy(harry_potter, P), likes_voldemort(X), enemy(P, george_weasley)]

```

239 :: 10 – 00:00:00 (453 nps)

Query 15: [enemy(_EX2, george_weasley)]
2 :: 1 – 00:00:00 (316 nps)

Query 16: [creature(P)]
2 :: 1 – 00:00:00 (142 nps)

Query 17: [in_gryffindor(_EX2)]
3 :: 2 – 00:00:00 (107 nps)

Query 18: [male(P), enemy(P, X), enemy(lucius_malfoy, X)]
925 :: 5 – 00:00:01 (887 nps)

Query 19: [enemy(draco_malfoy, P), enemy(bellatrix_lestrange, P)]
10 :: 4 – 00:00:00 (314 nps)

Query 20: [subject(P)]
2 :: 1 – 00:00:00 (143 nps)

Query 21: [in_gryffindor(_EX0)]
3 :: 2 – 00:00:00 (111 nps)

Query 22: [female(P), house(X, P), house(X, nymphadora_tonks), teaches(P, herbiology)]
5 :: 4 – 00:00:00 (56 nps)

Query 23: [enemy(X, _EX2)]
2 :: 1 – 00:00:00 (39 nps)

Query 24: [ancestor(arthur_weasley, Z)]
3 :: 2 – 00:00:00 (216 nps)

Query 25: [ancestor(harry_potter, _EX0)]
3 :: 2 – 00:00:00 (318 nps)

Query 26: [wizard(P)]
5 :: 3 – 00:00:00 (182 nps)

Query 27: [ancestor(harry_potter, P), descendant(P, lily_potter), female(P)]
10 :: 7 – 00:00:00 (49 nps)

Query 28: [enemy(sirius_black, _EX5)]
4 :: 3 – 00:00:00 (227 nps)

Query 29: [enemy(_EX4, peter_pettigrew)]
4 :: 3 – 00:00:00 (103 nps)

Query 30: [hates(Z, _EX2)]
5 :: 4 – 00:00:00 (216 nps)

Query 31: [enemy(molly_weasley, _EX3)]
18 :: 3 – 00:00:00 (490 nps)

Query 32: [male(P), enemy(P, X), enemy(X, Y), ancestor(Y, james_sirius_potter), parent(Y, lily_luna_potter)]

197 :: 8 – 00:00:00 (477 nps)

Query 33: [descendant(P, Y), descendant(P, X), enemy(X, bellatrix_lestrange), parent(Y, albus_severus_potter), ancestor(Y, P)]

111 :: 11 – 00:00:00 (162 nps)

Query 34: [in_gryffindor(P)]

3 :: 2 – 00:00:00 (104 nps)

Query 35: [enemy(X, seamus_finnigan)]

4 :: 3 – 00:00:00 (425 nps)

Query 36: [hates(P, X), hates(P, myrtle_warren), hates(bellatrix_lestrange, X)]

48 :: 9 – 00:00:00 (474 nps)

Query 37: [magical(P), creature(P), magical_creature(P)]

5 :: 4 – 00:00:00 (235 nps)

Query 38: [witch(lily_potter)]

6 :: 3 – 00:00:00 (617 nps)

Query 39: [hates(_EX3, petunia_dursley)]

14 :: 4 – 00:00:00 (632 nps)

Query 40: [male(P), ancestor(P, X), descendant(X, P), parent(P, X), wizard(X)]

9 :: 8 – 00:00:00 (36 nps)

Query 41: [house(X, P), in_hufflepuff(P), house(X, nymphadora_tonks)]

21 :: 4 – 00:00:00 (784 nps)

Query 42: [magical_creature(P), magical(P), creature(P)]

4 :: 3 – 00:00:00 (156 nps)

Query 43: [enemy(X, _EX5)]

2 :: 1 – 00:00:00 (39 nps)

Query 44: [magical_creature(X)]

4 :: 3 – 00:00:00 (142 nps)

Query 45: [descendant(_EX1, _EX3)]

6 :: 5 – 00:00:00 (93 nps)

Query 46: [hates(_EX4, _EX5)]

5 :: 4 – 00:00:00 (143 nps)

Query 47: [enemy(_EX5, neville_longbottom)]

2 :: 1 – 00:00:00 (195 nps)

Query 48: [descendant(_EX5, _EX2)]

4 :: 3 – 00:00:00 (69 nps)

Query 49: [descendant(james_sirius_potter, Z)]

221727 :: 5 – 00:00:08 (27378 nps)

Query 50: [betrays(P, ron_weasley), betrays(P, charlie_weasley)]
3 :: 2 - 00:00:00 (344 nps)

Query 51: [enemy(P, bellatrix_lestrange), betrays(peter_pettigrew, P)]
9 :: 4 - 00:00:00 (168 nps)

Query 52: [hates(Z, P)]
5 :: 4 - 00:00:00 (365 nps)

Query 53: [enemy(_EX2, _EX3)]
2 :: 1 - 00:00:00 (39 nps)

Query 54: [enemy(_EX3, neville_longbottom)]
2 :: 1 - 00:00:00 (337 nps)

Query 55: [horcrux(P), destroyed_horcrux(hermione_granger, P), destroyed(hermione_granger, P)]
4 :: 3 - 00:00:00 (519 nps)

Query 56: [wizard(P), descendant(P, X), enemy(X, ron_weasley)]
410 :: 9 - 00:00:00 (1431 nps)

Query 57: [subject(P)]
2 :: 1 - 00:00:00 (26408 nps)

Query 58: [hates(Z, X)]
5 :: 4 - 00:00:00 (831 nps)

Query 59: [enemy(_EX0, _EX3)]
2 :: 1 - 00:00:00 (43 nps)

Query 60: [enemy(P, ron_weasley), male(P)]
13 :: 2 - 00:00:00 (357 nps)

Query 61: [evil(Y)]
4 :: 3 - 00:00:00 (1558 nps)

Query 62: [hates(voldemort, P), hates(narcissa_malfoy, P)]
19 :: 8 - 00:00:00 (484 nps)

Query 63: [enemy(_EX4, _EX2)]
2 :: 1 - 00:00:00 (34 nps)

Query 64: [teaches(X, P), subject(P), professor(X)]
4 :: 3 - 00:00:00 (113 nps)

Query 65: [enemy(_EX1, hermione_granger)]
2 :: 1 - 00:00:00 (218 nps)

Query 66: [hates(_EX3, lily_potter)]
5 :: 4 - 00:00:00 (751 nps)

Query 67: [subject(P)]
2 :: 1 - 00:00:00 (34941 nps)

Query 68: [enemy(Y, severus_snape)]

2 :: 1 - 00:00:00 (284 nps)

Query 69: [house(P, newt_scamander), house(P, X), house(P, Y), teaches(X, herbology), male(Y)]

61 :: 5 - 00:00:00 (258 nps)

Query 70: [teaches(P, potions), house(X, P), house(X, regulus_black), male(P)]

20 :: 4 - 00:00:00 (814 nps)

Query 71: [descendant(X, _EX5)]

4 :: 3 - 00:00:00 (81 nps)

Query 72: [ancestor(arthur_weasley, _EX1)]

3 :: 2 - 00:00:00 (238 nps)

Query 73: [ancestor(james_potter, P), descendant(P, lily_potter), parent(ginny_weasley, P)]

3652 :: 9 - 00:00:00 (3747 nps)

Query 74: [creature(P)]

2 :: 1 - 00:00:00 (17907 nps)

Query 75: [hates(_EX3, _EX5)]

5 :: 4 - 00:00:00 (806 nps)

Query 76: [enemy(P, X)]

2 :: 1 - 00:00:00 (28242 nps)

Query 77: [in_slytherin(X)]

3 :: 2 - 00:00:00 (185 nps)

Query 78: [ancestor(_EX0, Y)]

3 :: 2 - 00:00:00 (64 nps)

Query 79: [enemy(_EX1, X)]

2 :: 1 - 00:00:00 (44 nps)

Query 80: [ancestor(_EX0, lily_luna_potter)]

362442 :: 4 - 00:00:34 (10412 nps)

Query 81: [wizard(_EX0)]

5 :: 3 - 00:00:00 (61 nps)

Query 82: [enemy(arthur_weasley, P), enemy(X, P), enemy(X, bellatrix_lestrange), member_of(P, death_eaters), member_of(X, order_of_the_phoenix)]

354 :: 9 - 00:00:00 (676 nps)

Query 83: [house(P, X), house(P, charlie_weasley), betrays(Y, X), enemy(harry_potter, Y), enemy(minerva_mccgonagall, Y)]

369 :: 8 - 00:00:00 (650 nps)

Query 84: [wizard(_EX2)]

6 :: 3 - 00:00:00 (81 nps)

Query 85: [enemy(_EX5, X)]

```

2 :: 1 - 00:00:00 (43 nps)

Query 86: [destroyed_horcrux(P, tom_riddles_diary), wizard(P), enemy(narcissa_malfoy, P)]
322 :: 9 - 00:00:00 (638 nps)

Query 87: [likes_voldemort(_EX5)]
4 :: 3 - 00:00:00 (297 nps)

Query 88: [hates(bellatrix_lestrange, _EX0)]
5 :: 4 - 00:00:00 (419 nps)

Query 89: [deatheater(_EX5)]
3 :: 2 - 00:00:00 (1586 nps)

Query 90: [descendant(_EX3, alice_longbottom)]
138221 :: 3 - 00:00:12 (10941 nps)

Query 91: [likes_voldemort(_EX4)]
4 :: 3 - 00:00:00 (2202 nps)

Query 92: [in_slytherin(narcissa_malfoy)]
3 :: 2 - 00:00:00 (1271 nps)

Query 93: [pure_blood(P)]
2 :: 1 - 00:00:00 (100 nps)

Query 94: [betrays(P, X), betrays(P, hermione_granger), betrays(percyy_weasley, X)]
4 :: 3 - 00:00:00 (51 nps)

Query 95: [descendant(dudley_dursley, petunia_dursley)]
4 :: 3 - 00:00:00 (466 nps)

Query 96: [in_hufflepuff(_EX0)]
3 :: 2 - 00:00:00 (368 nps)

Query 97: [hates(P, colin_creevey), enemy(X, P), descendant(Y, X), wizard(Y)]
16832 :: 13 - 00:00:01 (9187 nps)

Query 98: [subject(P)]
2 :: 1 - 00:00:00 (20250 nps)

Query 99: [teaches(P, Y), enemy(P, X), hates(X, petunia_dursley), teaches(gilderoy_lockhart, Y), pure_blood(X)]
401 :: 10 - 00:00:00 (1134 nps)

Query 100: [hates(peter_pettigrew, vernon_dursley)]
8 :: 4 - 00:00:00 (618 nps)

ming: 8782.98 nodes/query
Time to run all queries: 80.16418337399998

```

```

In [20]: !head std-46-110-3-50.csv
         !head ming-46-110-3-50.csv

```

```
query,std reasoner,std nodes explored,std min depth,success,time
1,std,6,5,True,0.0004671110000007417
2,std,4,3,True,0.000817774000000604
3,std,3,2,True,0.0002816600000006275
4,std,4,3,True,0.0004513309999998327
5,std,4,3,True,0.00035616899999979523
6,std,5,4,True,0.0003641789999999645
7,std,2,1,True,0.00043512000000056616
8,std,4,3,True,0.0003254919999999828
9,std,2,1,True,0.0002981450000003605
query,ming reasoner,ming nodes explored,ming min depth,success,time
1,ming,10,3,True,0.5301936640000005
2,ming,18,3,True,0.05737685899999967
3,ming,3,2,True,0.005274327000000412
4,ming,4,3,True,0.08509362500000073
5,ming,6,5,True,0.17237762799999956
6,ming,5,4,True,0.025274366999999742
7,ming,2,1,True,0.08079707599999963
8,ming,4,3,True,0.013159890000000729
9,ming,2,1,True,0.009904683000000247
```